



FACULTY OF ENGINEERING AND SCIENCE
SCHOOL OF ENGINEERING

PROJECT REPORT
2024/2025

**Design of an alternative transportation method in the
context of the “Ruta de los Volcanes” visit in the
Timanfaya National Park**

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Engineering, Design, and Innovation BEng (Hons)

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Abstract

This project designed a sustainable electric shuttle system for Timanfaya National Park's "Ruta de los Volcanes." By substituting solar-powered electric shuttles for diesel-powered buses, the project aimed to lower carbon emissions, improve tourist satisfaction, and guarantee long-term operational sustainability. The research enhanced accessibility for guests while supporting the park's commitment to environmental preservation by utilizing renewable energy sources and innovative vehicle design, in order to maintain the optimal balance between operational effectiveness and ecological preservation.

The research used a design-led approach, following the Double Diamond framework. Computational modelling, CAD-based design, and energy consumption simulations were key elements of the process. By applying universal design principles, the final shuttle concept ensured inclusivity by accommodating up to 12 people and two wheelchair users.

Compared to the current diesel system, performance research shows that the proposed system can eliminate about 40,000 kg of CO₂ per shuttle year, resulting in a 100% reduction in direct CO₂ emissions. According to energy calculations, a separate solar charging infrastructure is needed to support each shuttle's 6.18 kWh every tour. A fleet of 20 shuttles is advised by operational capacity study to manage daily visitor loads of up to 1,000. The results will provide guidance for future studies on eco-friendly mobility options and sustainable tourism, and they will act as a model for deploying renewable energy-powered transportation in protected regions.

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List of Abbreviations and Mathematical Symbols

List of Mathematical Symbols

Symbol	Definition	Unit
m	Vehicle Mass	kg
g	Gravitational Acceleration (9.81)	m/s ²
d	Distance Travelled (Circuit Length)	m
Cr	Rolling Resistance Coefficient	-
h	Elevation Gain	m
Pacc	Accessory Load Power	kW
t	Time of Tour	h
Err	Rolling Resistance Energy	MJ
Eelev	Elevation Energy	MJ
Eacc	Accessory Load Energy	MJ
Etotel	Total Energy Consumption per Tour	MJ / kWh

Table 1: List of Math Symbols

List of Abbreviations

Abbreviation	Definition
ADA	Americans with Disabilities Act (Accessibility Standard)
CAD	Computer-Aided Design
CO ₂	Carbon Dioxide

Abbreviation	Definition
Cr	Rolling Resistance Coefficient
EV	Electric Vehicle
GFRP	Glass Fibre Reinforced Polymer
kWh	Kilowatt Hour
LFP	Lithium Iron Phosphate (Battery Type)
MJ	Megajoule
PV	Photovoltaic (Solar Panel)
SDG	Sustainable Development Goals
UNESCO	United Nations Educational, Scientific and Cultural Organization

Table 2: List of Abbreviations

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1. INTRODUCTION

1.1 Background of Project

Famous for its distinctive volcanic terrain, Timanfaya National Park is a UNESCO Biosphere Reserve situated in Lanzarote, Canary Islands (UNESCO, 2023). The 51-square-kilometer park is one of Spain's most frequented protected places, drawing over 1.5 million visitors a year (López, 2024). A major objective is minimizing human damage while preserving an interesting tourist experience because of its fragile volcanic ecosystem.

At the moment, diesel-powered buses are used for guest transportation, which results in CO₂ emissions, noise pollution, and environmental damage (IEA, 2023). According to research, conventional diesel buses cause serious air quality issues in conservation zones since they release about 2.7 kg of CO₂ for every liter of fuel used. Furthermore, visitors with mobility disabilities are unable to fully experience the park's landscape due to its restricted accessibility choices (IEA, 2023).

Many national parks have started using electric or solar-powered shuttle systems to lessen their environmental effect as a result of the global movement towards sustainable transportation (Gilleran et al., 2015). According to case studies from Yosemite and Zion National Parks, electric shuttles reduce emissions by more than 90% when compared to diesel buses, improving the visitor experience while preserving ecological integrity (Steinbach, 2023). Motivated by these achievements, this project investigates the viability of installing an electric shuttle system that runs on solar power at Timanfaya National Park. This would guarantee zero direct CO₂ emissions, enhanced accessibility, and long-term operational sustainability.



Figure 1: Ruta de los Volcanes ((Volcano Route), 2021)

1.2 Aims and Objectives of the Project

1.2.1 Project Aims

The project's aim is to create a solar-powered electric shuttle system that will improve accessibility, remove direct CO₂ emissions, and improve the experience of the Timanfaya National Park.

1.2.2 Project Objectives

1- *Protecting the Environment*

- Implementation of Electric Shuttles: To reduce emissions and improve sustainability, entirely electric shuttles will be used.

2- *Enhanced Visitor Experience*

- Improved Services: The new electric buses provide air conditioning and quieter operations, which improves the experience of visitors.

3- *Sustainability and Viability*

- Sustainable practices and increased energy efficiency are two benefits of using lightweight, environmentally friendly materials in shuttle design.
- Infrastructure for Solar Charging: Lanzarote is ideally situated for the use of solar energy because it receives roughly 3,010 hours of sunshine every year. (Lanzarote climate: weather by month, temperature, rain - Climates to Travel, no date)

4- Accessibility

- Better adaptations, such as air conditioning and other accessibility features, are included in the inclusive new electric shuttle design.
- Following the universal design principles improves inclusion by guaranteeing accessibility for all guests.

1.3 Scope of the Project

Sustainable and accessible electric shuttle system design which runs on solar power for Timanfaya National Park is the main objective of this project. While omitting outside variables like shifting governmental regulations and tourism management, the scope covers important topics like vehicle design, energy infrastructure, accessibility, and sustainability.

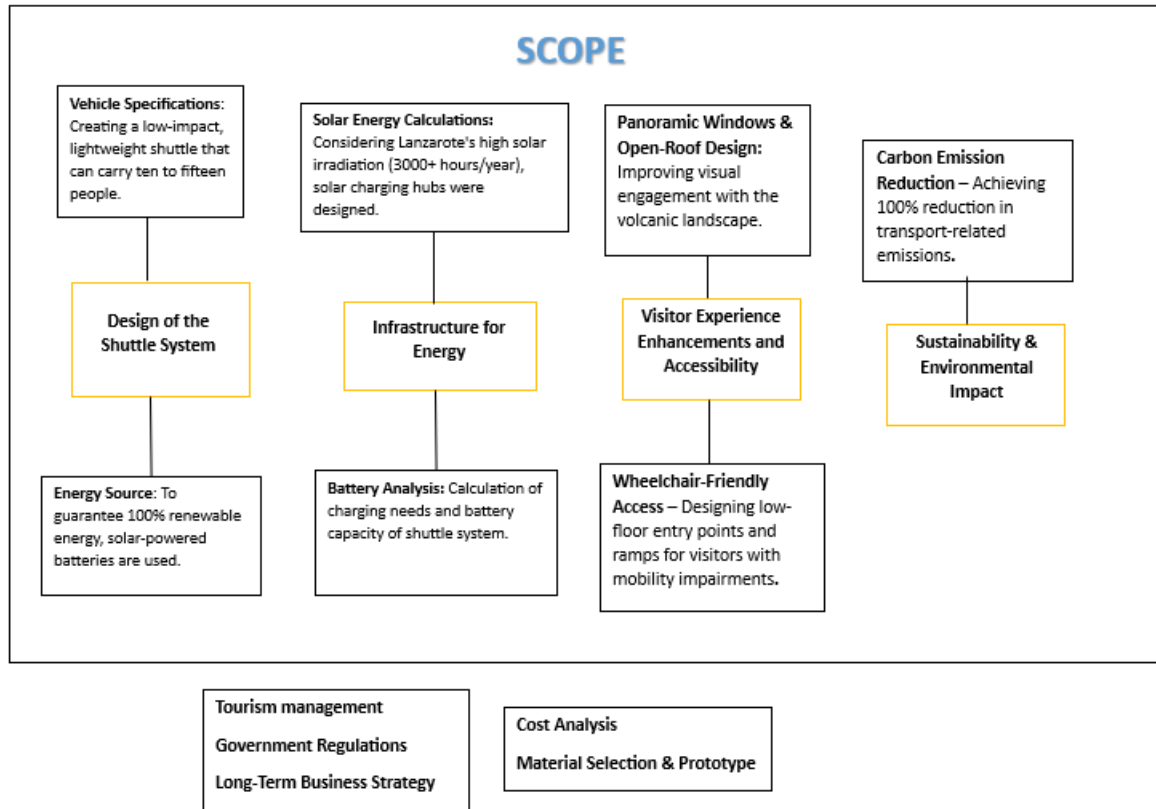


Figure 2: Scope Diagram for The Project

1.4 Report Outline

There are multiple chapters in this report. The project's background, aims, objectives, and scope were presented in Chapter 1. The knowledge base for our design is established by Chapter 2 (Literature Review), which examines relevant articles and earlier projects in fields such as inclusive design, electric vehicle technology and solar charging, ecotourism visitor experience, and sustainable transportation in protected areas. The research and design process, including the Double Diamond framework used to identify user needs, specify requirements, create solutions (using CAD modelling and analysis), and present the finished proposal, is covered in Chapter 3 (Methodology). The materials and techniques used in the shuttle system's design and

evaluation are also covered. The project's main outputs are shown in Chapter 4 (Results), which includes the suggested shuttle design (complete with engineering drawings and CAD renders), performance simulations (energy consumption, charging requirements). These findings are interpreted in Chapter 5 (Discussion), which also discusses the design's practical consequences for Timanfaya National Park and how well it satisfies the project's goals. The findings are compiled in Chapter 6 (Conclusions and Future Work), which also emphasizes the project's contributions and makes recommendations for future improvements or implementation steps. All sources are cited, and supporting documentation, including the project Gantt chart, is included in an appendix.



Figure 3: Ruta de los volcanes 2 (Ruta de los Volcanes en La Palma: Guía y consejos prácticos, 2020)

2. LITERATURE REVIEW

2.1Introduction

This chapter critically examines the body of research and case studies on solar energy integration, tourist experience, accessibility, and sustainable transportation options in protected

areas. The purpose of the review is to determine the gaps in knowledge that this project fills and to set the scene for the proposed shuttle system in Timanfaya National Park.

Heading 1	Heading 2	Heading 3	Heading 4
Sustainable Transportation in Protected Areas	Electric Vehicles and Solar Energy Integration	Visitor Experience and Interpretive Value	Accessibility and Inclusive Design

Table 3: Literature Review Topics

2.2Theme 1: Sustainable Transportation in Protected Areas

Several studies demonstrate how traditional diesel transport methods in protected nature regions have a negative environmental impact. Conventional automobiles release pollutants that harm nearby ecosystems and lower air quality (Garcia & Martinez, 2021). Yosemite and Zion National Parks case studies show that switching to electric or hybrid shuttle systems enhances the visitor experience while reducing emissions by more than 90% (Steinbach, 2023).

Small island contexts like Lanzarote, where solar energy potential is higher but infrastructure constraints differ, have received less attention than parks in North America, where the majority of study is concentrated. Additionally, prior deployments have documented issues with battery capacity and maintenance expenses (NREL, 2022), necessitating more creative and situation-specific solutions.



Figure 4: Ruta de los volcanes 4(Ruta de los Volcanes (Volcano Route), 2021)

2.3Theme 2: Solar Energy Integration

The use of solar energy in transportation has become more popular worldwide. The viability of putting removable solar panels in between railway tracks has been shown by initiatives like Sun-Ways in Switzerland (Sun-Ways, 2023). In a comparable way, the SolaRoad project in the Netherlands and India's railway solarization demonstrate the possibility of fusing energy generation with mobility.

Despite their achievements, these initiatives have important drawbacks. For example, the Wattway project in France has problems with durability and excessive costs (Wattway, 2022). Research on the long-term viability of solar infrastructure in isolated and ecologically delicate locations is lacking. By assessing the technological and economic feasibility of solar-powered shuttles designed especially for Timanfaya National Park, this project fills this gap.

2.4 Theme 3: Visitor Experience and Interpretive Value

Studies on tourism indicate that comfort, accessibility, and educational value all affect how satisfied visitors are in protected areas (UNWTO, 2022). Due to noise, vibration, and enclosed spaces that limit views and interpretation, Timanfaya's diesel-powered buses discourage tourist interaction.

According to Gilleran et al. (2015), open-sided electric tram designs increase tourist happiness and immersion. However, a thorough examination of how these mobility networks can improve interpretive experiences in volcanic environments in particular is lacking in the research now in publication. By using audio guides and design elements specific to Timanfaya's volcanic topography, this initiative aims to close this gap.



Figure 5: Ruta de los volcanes (<https://www.facebook.com/mablanquer>, 2019)

2.5 Theme 4: Accessibility and Inclusive Design

Concern over accessibility in public transport is growing, especially in tourist environments. According to Rodriguez et al. (2022), inaccessible vehicles and the shortage of interpretive resources are two obstacles that visitors with mobility impairments encounter in natural parks. Wheelchair space, railings, and low-floor vehicles are among the universal design concepts that are promoted by international guidelines (Department for Transport UK, 2020).

Studies that have already been done mostly concentrate on urban transit, paying little consideration to accessibility in ecotourism settings. Applying inclusive design to shuttle systems in protected volcanic areas is an area that needs more study. In order to meet this demand, the shuttle idea incorporates universal design elements.

2.6 Summary and Identified Gaps

The literature shows how sustainable mobility systems in protected regions have positive social and environmental effects. But it also highlights a number of gaps:

- Sustainable shuttle systems in small island biosphere reserves are not getting a lot of focus.
- There are few studies on the economic viability of solar-powered transportation infrastructure.
- There aren't many studies that comprehensively combine sustainability, accessibility, and visitor experience in volcanic settings.

By creating a solar-powered electric shuttle system that takes into account accessibility, visitor enjoyment, and environmental preservation in Timanfaya National Park, this project seeks to close these gaps.

3. METHODOLOGY

3.1 Research Methodology

The Double Diamond Framework, developed by the UK Design Council, served as the foundation for this project's design-led research process. Discover, Define, Develop, and Deliver are the four main stages that the framework uses to facilitate iterative development.

A survey of the literature and an analysis of sustainable transport systems' best practices were carried out during the Discover phase. Additional stakeholder inputs were obtained through evaluations of visitor reports and casual conversations with park employees.

The project's goals, parameters, and technical specifications were determined during the Define phase. "How might we design a zero-emission, accessible, and visitor-friendly shuttle system for Timanfaya National Park?" was the problem statement that was developed.

Concept creation and technical assessment were the main goals of the Develop phase. Using CAD modelling and hand sketching, a number of design options were investigated. System performance was evaluated by capacity analysis and energy simulations.

The best design was ultimately chosen during the Deliver phase, and the results were combined to create this report.

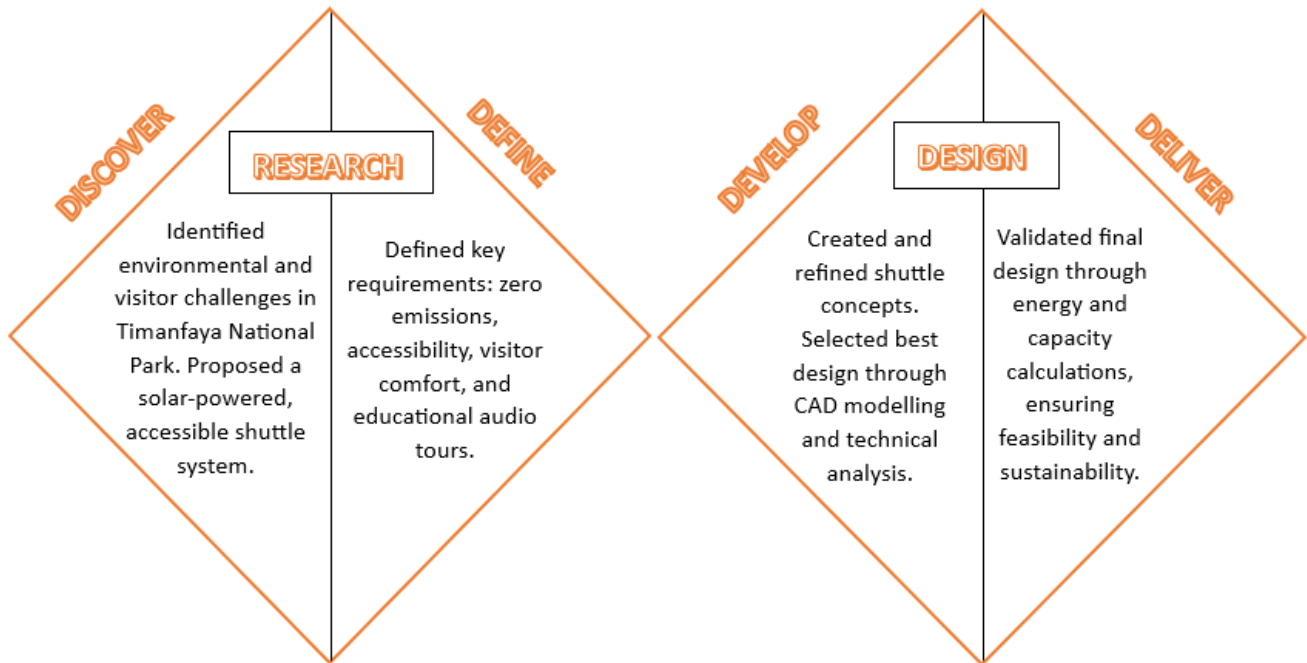


Figure 6: Double Diamond Workflow

3.2 Materials

No actual prototype was created because the project was a virtual design study. The following equipment and supplies were used:

- Software: Microsoft Excel for computations and capacity analysis; SolidWorks for CAD modelling.
- Materials (Theoretical): Design includes lithium-ion batteries, monocrystalline solar panels, composite panels for the body, lightweight aluminum alloy for the chassis, and laminated safety glass for the windows. Durability, cost-effectiveness, and strength-to-weight ratio were given first priority when choosing materials.

3.3 Methods

The project used a mixed-methods engineering approach, combining quantitative analysis techniques with qualitative design techniques. An overview of the main techniques employed is provided below:

- *CAD modelling and Design Engineering:*

Iterative engineering design was the main technique. The project started by developing basic sketches and gradually progressed to precise CAD drawings and 3D models, adhering to traditional design processes. The CAD model was used to virtually test component fittings such as the spatial arrangement of seats, batteries, and other internal features—thereby verifying compliance with geometric and ergonomic specifications. For example, the model confirmed that the floor height could be maintained below 300 mm, in accordance with ADA regulations, to permit ramp access. The CAD environment was also utilized to assess weight distribution and balance by allocating various materials to different components. This enabled the calculation of the shuttle's center of gravity and its stability on sloped terrain.

- *Analysis of Case Studies:*

During the Discover phase of the project, Case studies of other parks that had implemented electric or hybrid shuttle systems—such as Yosemite National Park and Zion National Park—were examined. Drawing on information from reports and published articles, this qualitative analysis identified key outcomes, challenges, and best practices. The findings were systematically recorded and integrated into the project's design specifications. For instance, insights from Zion's experience highlighted the need for effective battery cooling in hot climates, which was addressed in the design by incorporating ventilation into the system.

- *Computational Simulations:*

A number of analytical computations and basic simulations were conducted as part of the project:

Energy Simulation: The shuttle's energy consumption was estimated using the "Ruta de los Volcanes" circuit. Calculations considered topographical data (elevation changes) and a conservative driving cycle, including frequent stops and a low average speed (~20 km/h). Additional estimations included accessory loads such as the audio system and cooling fans, rolling resistance, and gravitational effort on inclines. Based on these factors, the required battery size was determined, with results indicating that a 20 kWh battery would support two to three complete circuits before requiring recharging.

Solar Panel Output Calculation: The potential daily electricity production of a designated solar panel area was calculated using average solar radiation data for Lanzarote obtained from climatological sources. It was determined that approximately 200 high-efficiency panels could generate over 600 kWh on a sunny day—sufficient to recharge the batteries of the entire shuttle fleet under most conditions.

Visitor–Capacity Calculation: Each electric shuttle is capable of accommodating 12 to 13 passengers, whereas the existing diesel buses carry approximately 50 passengers. Based on park visitation data (with peaks of up to ~1000 visitors per day), a fleet of 20 electric shuttles was calculated to be sufficient for managing visitor flow during high-demand periods. Detailed computations are provided in the Results section.

In conclusion, the methodology integrated evidence-based analysis with creative design strategies. The project remained iterative and user-centered through the application of the Double Diamond design framework.

4. RESULTS

The suggested shuttle system architecture and related studies are included in this section. The design of the solar charging infrastructure, the conceptual design of the electric shuttle (including CAD drawings and renderings), and quantitative assessments of the system's performance (emissions avoided, energy consumption, etc.) are important deliverables. The design and results are shown using figures and tables.

4.1 CAD Drawings

4.1.1 Early Concepts and Sketches:

Creating conceptual models that prioritised guest comfort, accessibility, sustainability, and operating efficiency was the main goal of the shuttle design's early development stage. Using CAD software and hand-drawn layout sketches, multiple design iterations were produced in order to assess passenger flow and space usage.



Figure 7: Concept 1.1

One of the earliest design proposals is depicted in Figures, which is distinguished by an open side window, a transparent sloping roof.

This concept's main goal was to maximize visibility of the volcanic terrain in order to provide an open and immersive tourist experience. In order to improve picturesque views, seats were positioned along the sidewalls, allowing passengers to view outside.

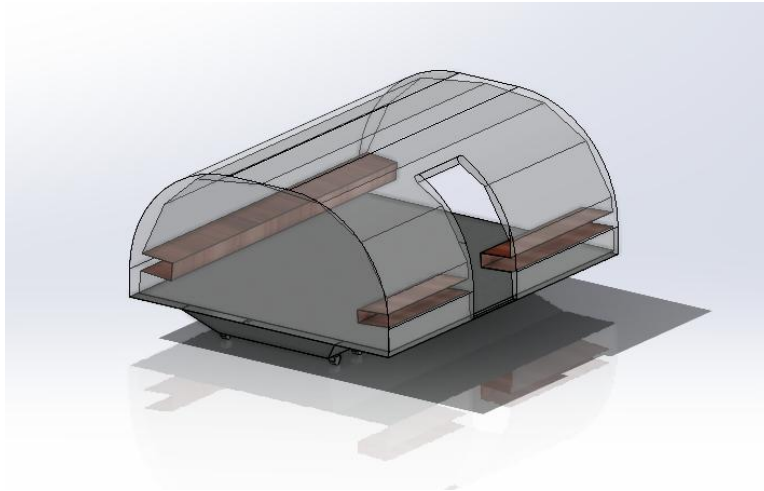


Figure 8: Concep1.2

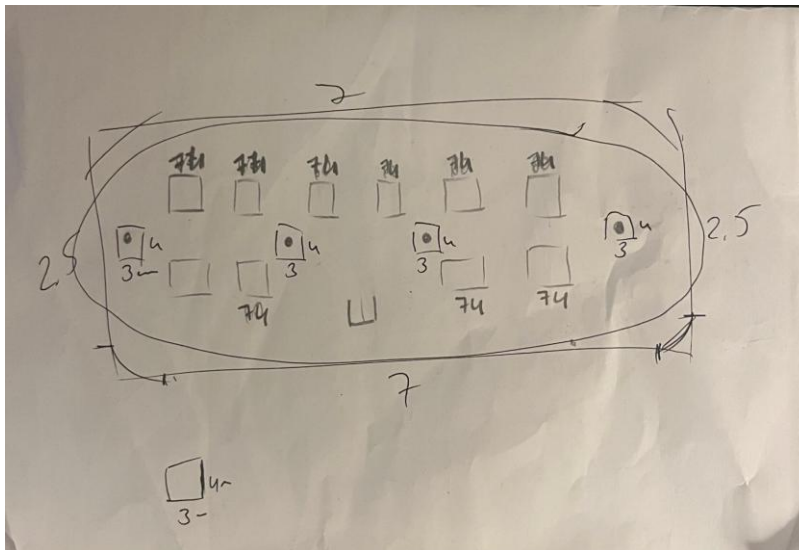


Figure 9: Hand - Drawn Sketch

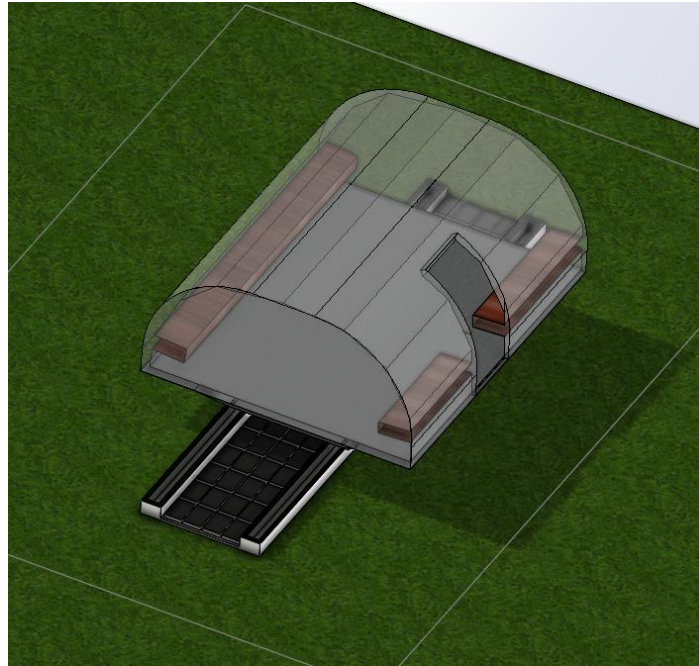


Figure 10: Concept 1.3

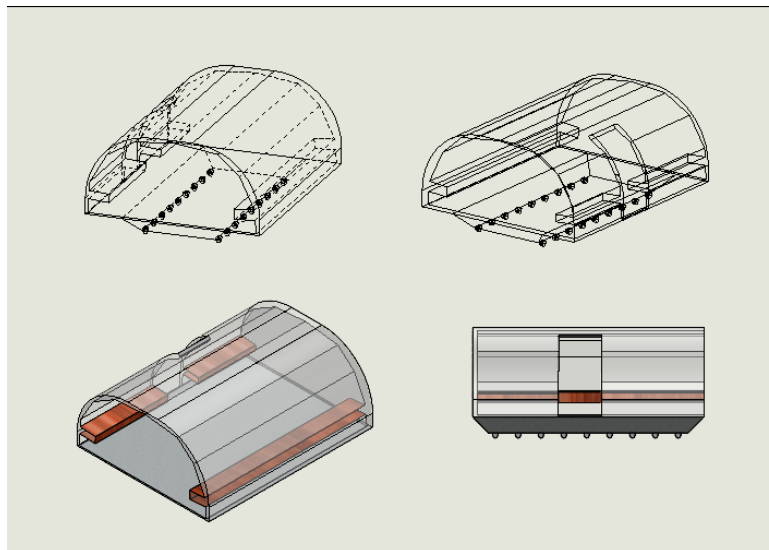


Figure 11: Drawing of Concept 1

The early concept development phase enabled the identification of key design criteria:

- Maximise visibility for visitors.
- Provide safe and easy access for wheelchair users.

- Minimise structural complexity while ensuring durability.
- Optimise internal space for comfort and capacity.

The iterative design process informed the development of the final shuttle concept presented in the next section

4.1.2 Final Concept:

The final shuttle design is a 10-seater electric shuttle bus that can accommodate two wheelchair users and is best suited for touring Timanfaya's landscape. About 8 meters long, 3.5 meters wide, and 4 meters tall, the vehicle is comparable to a large city tram system. For easy boarding, it has an open doorway in the center and two rows of forward-facing bench seats. The vehicle's central line has four handrails to improve visibility. The shuttle's rear axle is powered by an electric motor that gets its power from a lithium-ion battery pack.

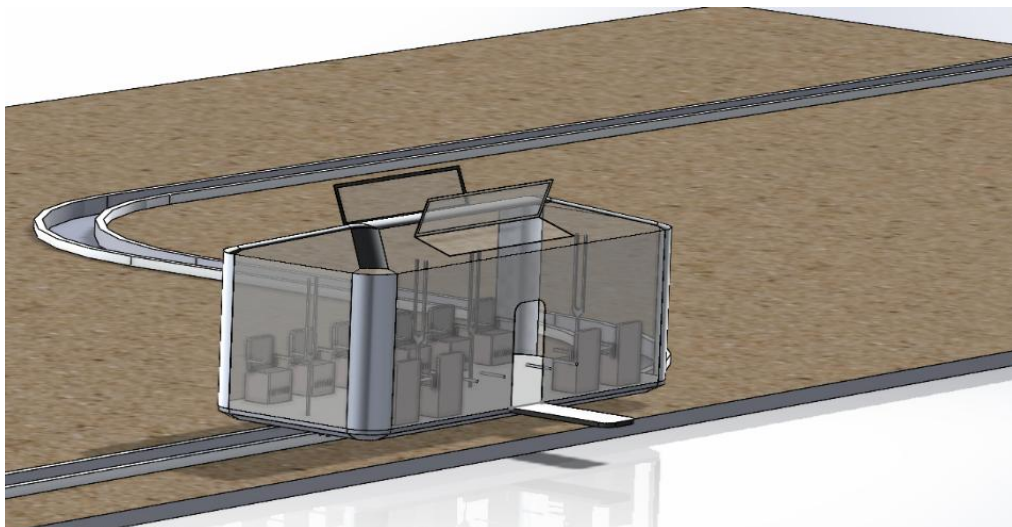


Figure 12 : CAD Model 1

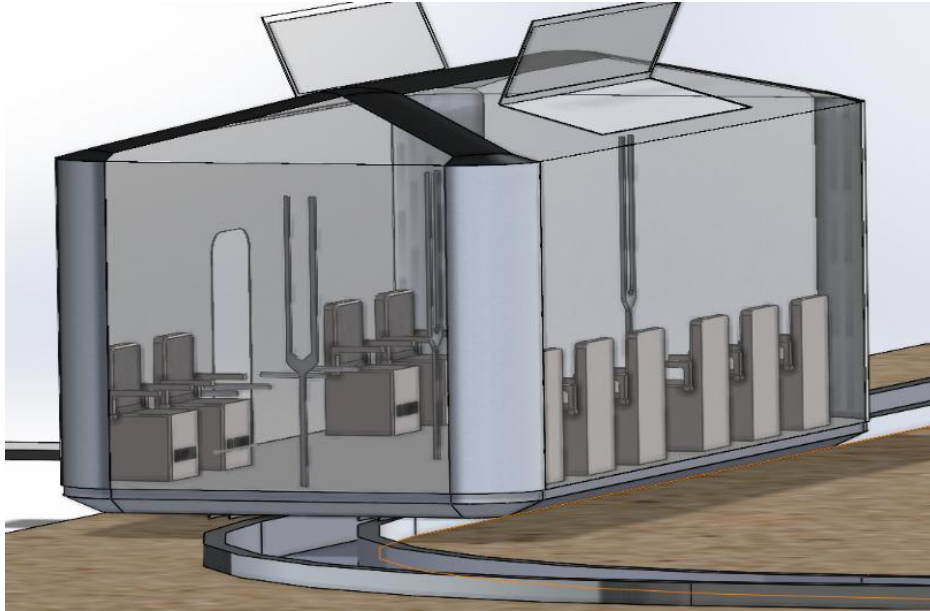


Figure 13: CAD Model 2

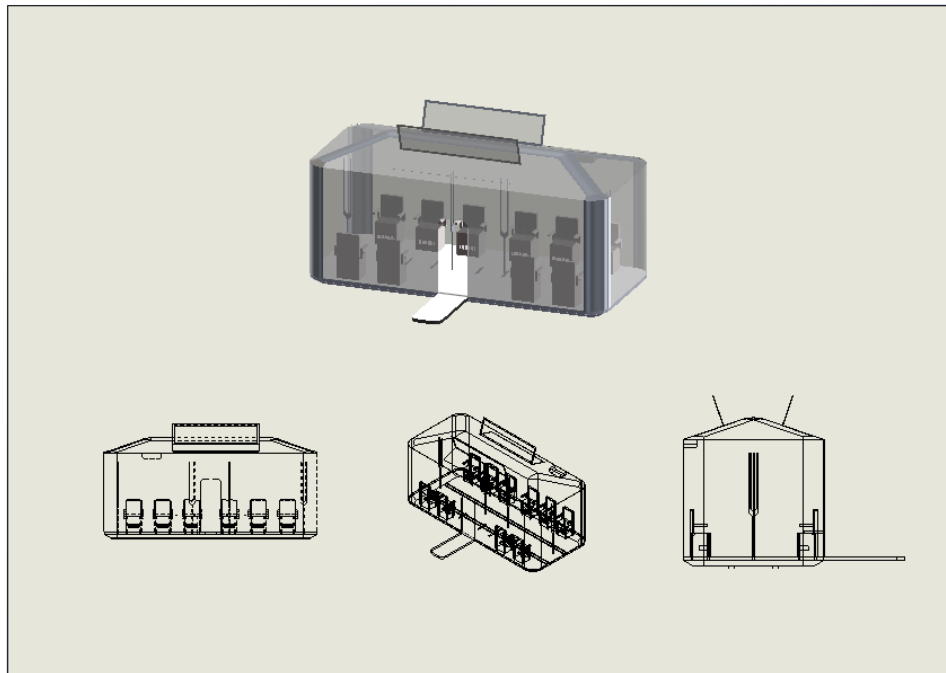


Figure 14: Drawing 1

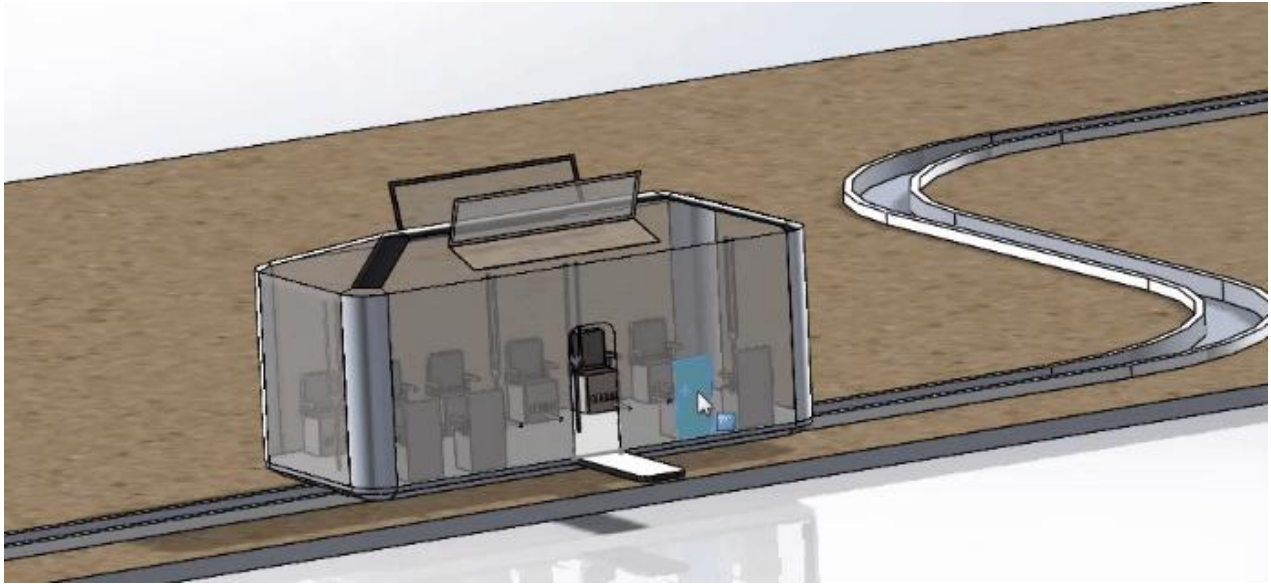


Figure 15: Simulation 1

Parameter	Value
Length	8.0 m
Width	3.5 m
Height	4.0 m
Passenger Capacity	12 (including 2 wheelchair users)
Battery Capacity	20 kWh
Drive System	72V Electric Motor

Table 4: Shuttle Specifications

4.2 Accessibility Features

The shuttle has a number of accessible features:

- Handrails and Support Bars: For passenger safety they are positioned close to seats and entrances (Figure 12)
- Wheelchair Ramp: Enables guests with limited mobility to board safely (Figure 11,13)
- Audio System: GPS-activated, multilingual guide with interpretive value. (Figure 15)

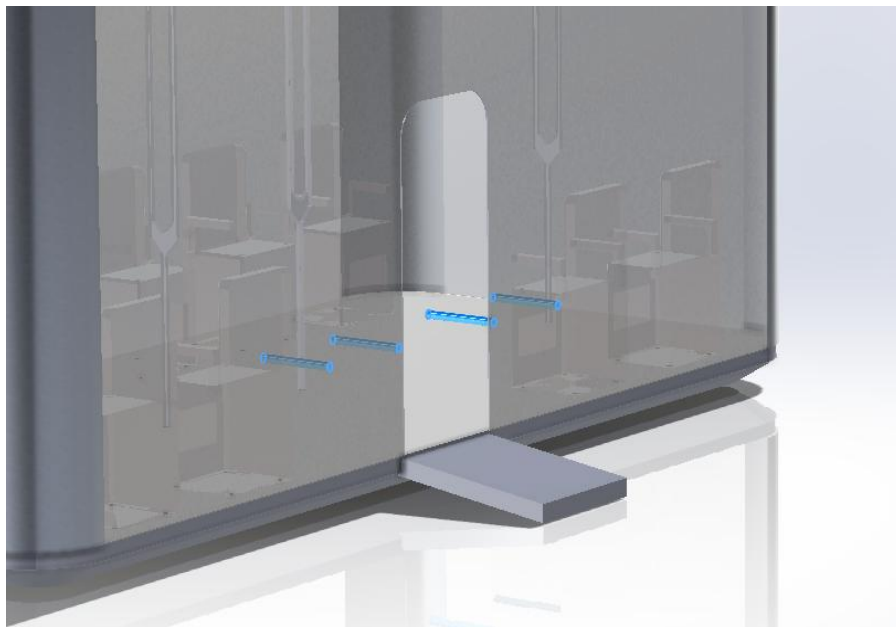


Figure 16: Wheelchair Ramp

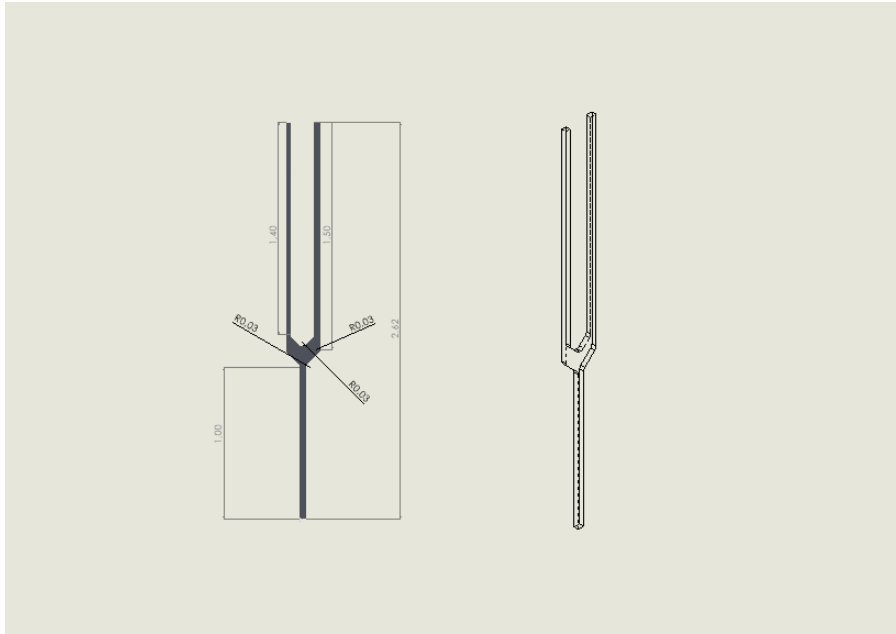


Figure 17: Handrails

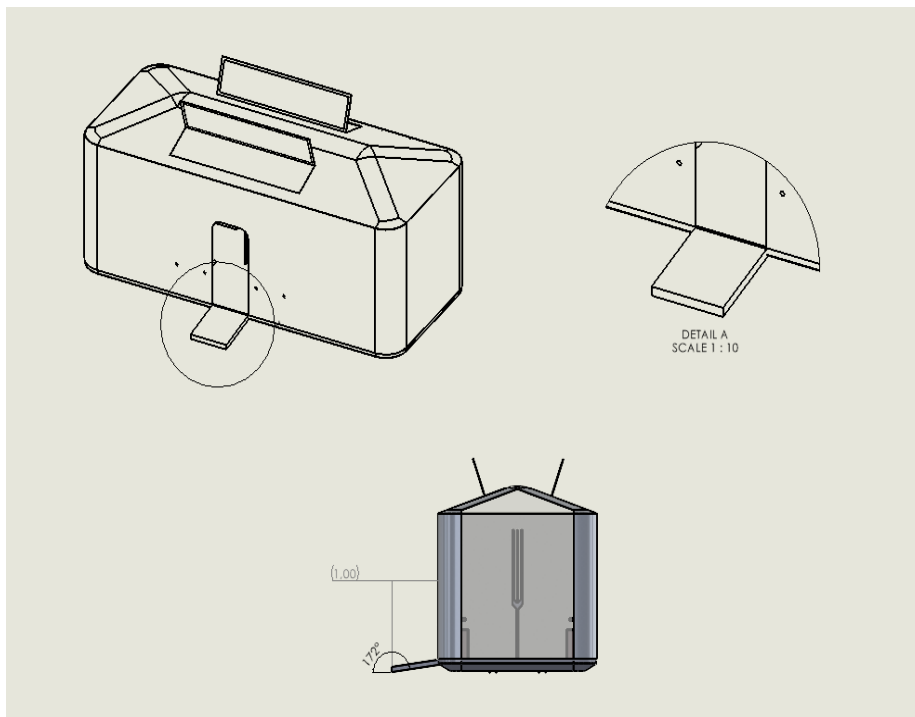


Figure 18: Wheelchair Ramp 2

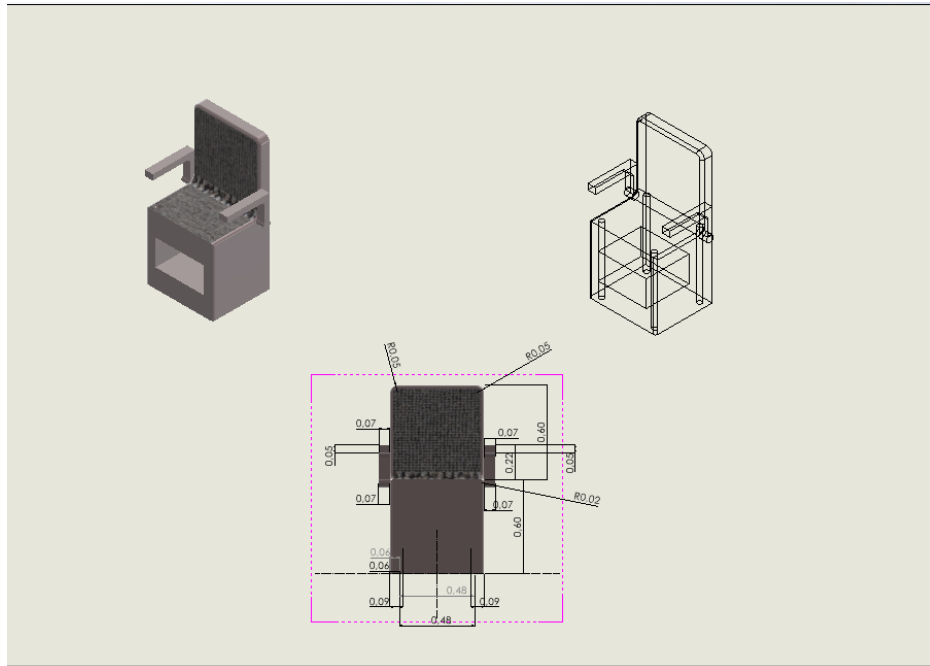


Figure 19: Seats

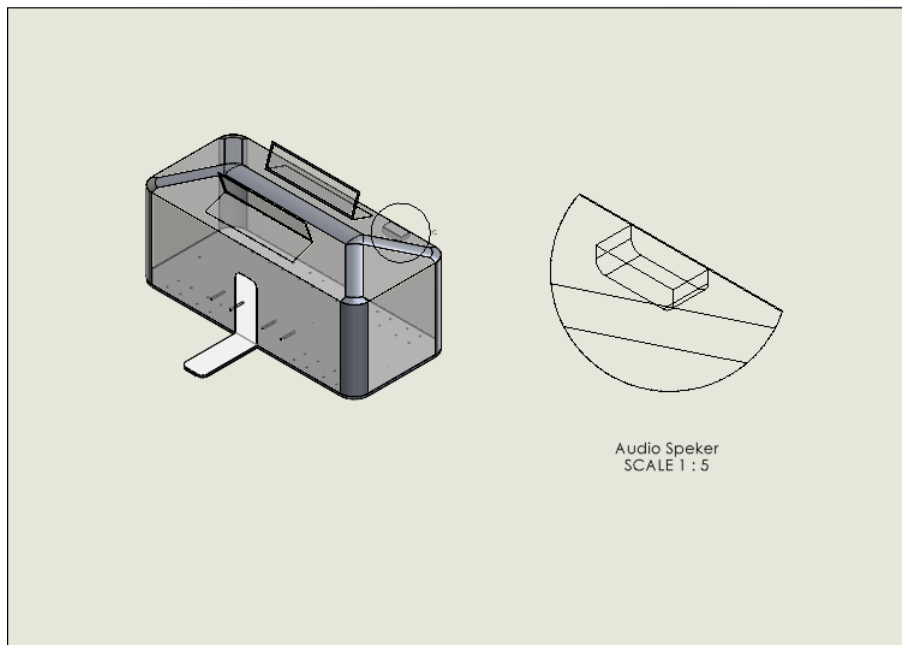


Figure 20: Audio Speaker

4.3 Solar Charging Infrastructure

In order to preserve the landscape view, the energy system is based on stationary solar panels that are positioned in between tramway rails. The table provides a summary of the specs.

Parameter	Value
Panel Area	2 m ² per panel
Efficiency	20%
Daily Solar Irradiance	8 hours/day
Daily Energy Production	~3.2 kWh/panel
Total Panels Required	167

Table 5: Solar Panel Specifications

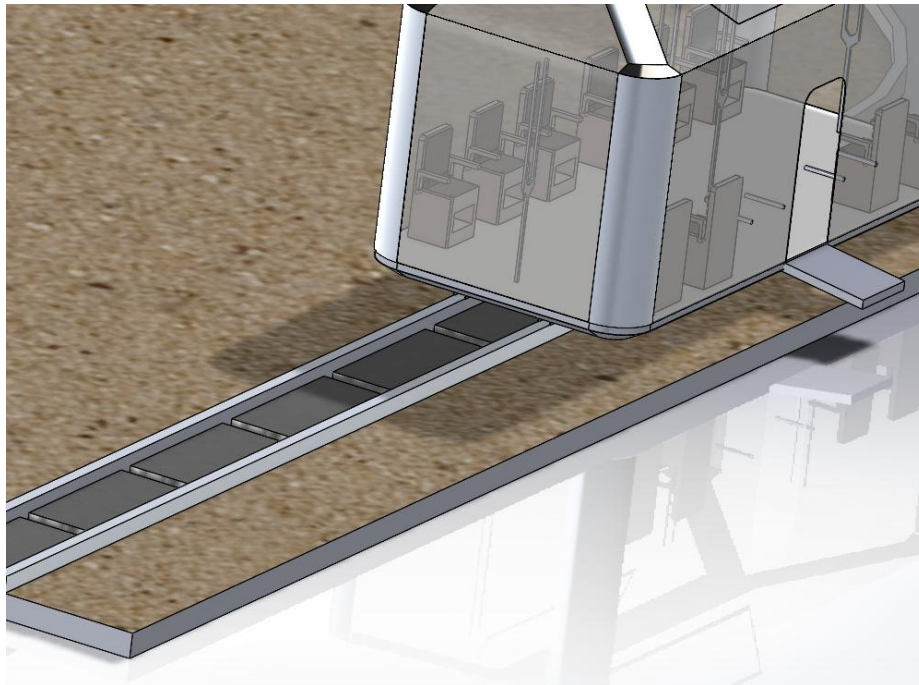


Figure 21: Solar Pannels

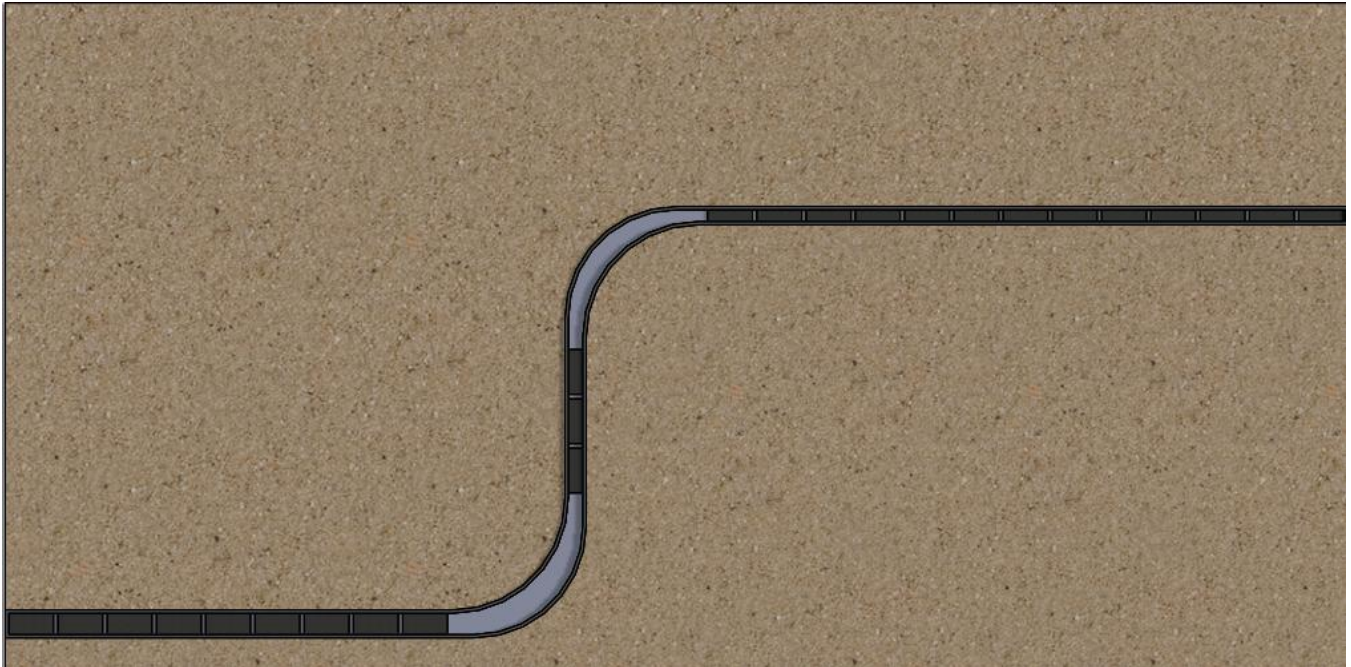


Figure 22: Solar Pannels 2

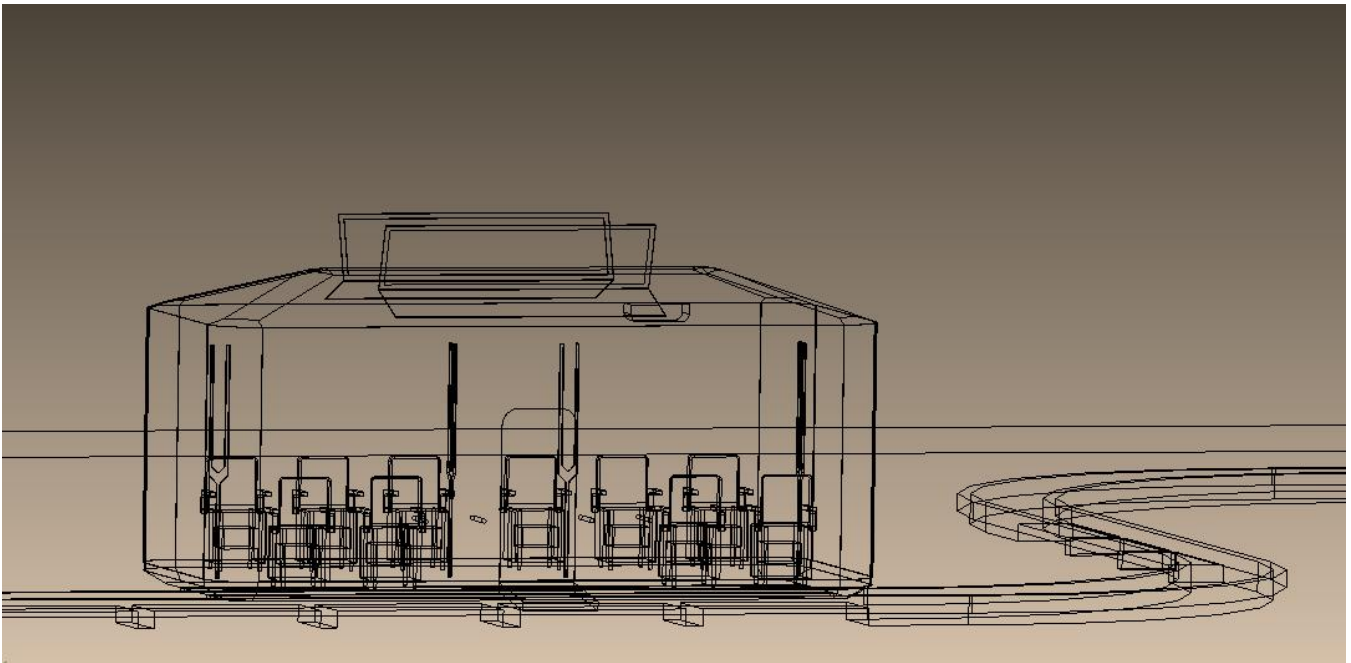


Figure 23: Vehicle Drawing

4.4 Calculations:

- Emissions Reduction:

There are no tailpipe emissions from the shuttle itself. Each electric shuttle will reduce emissions on each journey by replacing the current diesel coaches, which are anticipated to emit about 27 kg of CO₂ per tour, based on route length and diesel use. Assuming five shuttle journeys a day for 300 days, this amounts to more than 40,000 kg of CO₂ being removed annually per shuttle over the course of a year.

One vehicle's carbon emission per tour x

Number of vehicles x

Number of tour days (annually)

= Total Carbon Emission Per Year

Tens of thousands of kilograms of CO₂ can be removed by the park with several shuttles. When it comes to noise, EV motors are almost silent, so guests on board and nearby will hear significantly less noise than diesel engines. This improves the environment and the experience for tourists and helps wildlife by lowering noise pollution.

- Capacity and Throughput:

The present diesel bus can accommodate about 50 passengers, however each shuttle can accommodate up to 12–13 passengers. An optimal fleet size of 20 shuttles were determined to handle peak loads (based on park statistics of up to ~1000 visitors per day). Several shuttles would run on a varied schedule in order to sustain capacity (Timanfaya

frequently receives dozens of people every hour). For instance, twenty shuttles may depart continually, leaving every 5-6 minutes. According to scheduling research, a fleet of 20 could meet demand by making about 50 trips a day, even after calculating the charging times.

Daily visitors: 1,000

Number of shuttles: 20

Shuttle capacity: 12 passengers per shuttle

Operating hours: Assume 8 hours/day (typical working hours)

Calculate how many tours per day are needed for 1,000 visitors:

1

$$\frac{1,000 \text{ visitors/day}}{12 \text{ passengers/tour}} \approx 84 \text{ tours/day}$$

Operating period per day: 8 hours = 480 minutes

Total tours needed per day: 84 tours

Departure frequency calculation:

2

$$\frac{480 \text{ minutes}}{84 \text{ tours}} \approx 5.7 \text{ minutes/tour}$$

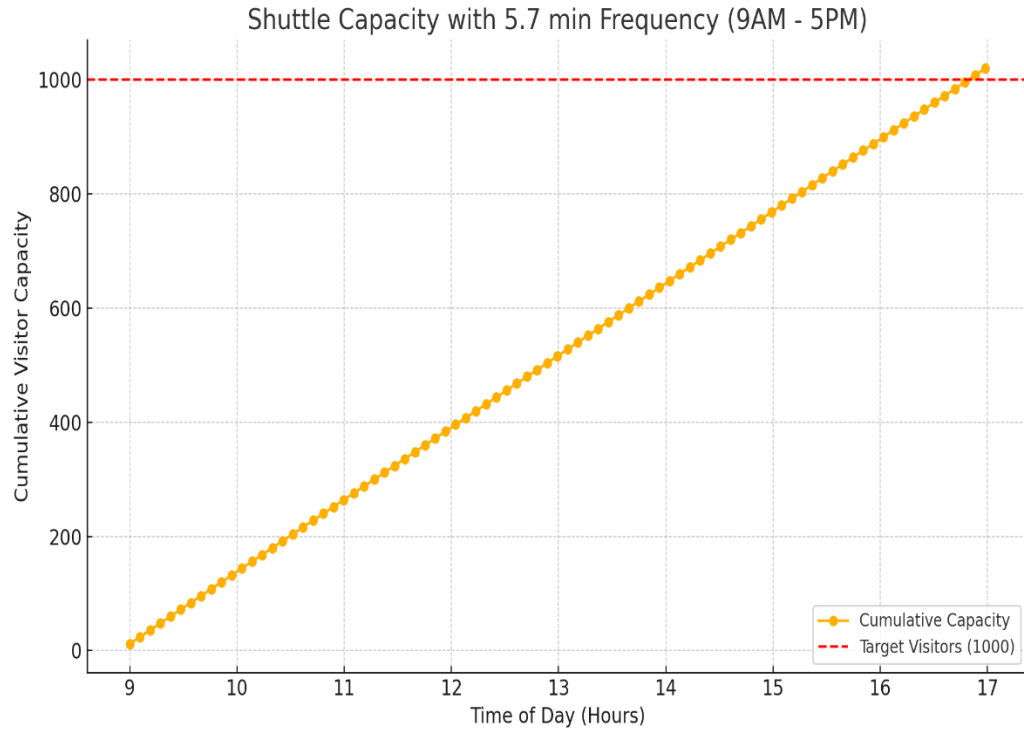


Figure 24: Frequency Graph

- Energy Efficiency:

According to the energy use simulation, each Ruta de los Volcanes loop uses about 5 kWh of battery power. Using 5 kWh per loop with a 20 kWh battery provides a comfortable safety factor. On most sunny days, solar power alone can sustain many tours because the solar station's output of 50 kWh per day can theoretically recharge the shuttle two to three times over. The system may require a tiny amount of grid electricity on rare days with cloudy sky, but overall, the design satisfies the sustainability goal of operating normally without the use of fossil fuels.

Battery Capacity: 20 kWh per shuttle

Daily Energy Consumption per Shuttle: approximately 20 kWh/day (assuming one full recharge per shuttle per day).

If running multiple shuttles (e.g., 4–5), total daily energy required would be around:

$$25.95 \text{ kWh (per shuttle)} \times 20 \text{ shuttles} = 519 \text{ kWh/day}$$

1

Dimensions: $1 \text{ m}^2 \times 2 \text{ m}^2 = 2 \text{ m}^2$

Efficiency: 20% (typical for standard panels)

Solar irradiance: 1 kW/m (standard average)

Power Output (kW) per panel

Each square meter gives 0.2kW at 20% efficiency.

Therefore, one panel (2 m^2):

$$2 \text{ m}^2 \times 0.2 \text{ kW/m}^2 = 0.4 \text{ kW} = 400 \text{ W}$$

2

Daily Energy Production (kWh/day)

*Average sunlight in Lanzarote: **8 hours/day***

Daily energy from one panel:

$$0.4 \text{ kW} \times 8 \text{ hours} = 3.2 \text{ kWh/day}$$

3

Therefore, to produce 400 kWh/day energy, needed solar panels:

$$\frac{519 \pm 10 \text{ kWh/day}}{3.2 \text{ kWh/day}} \approx 162 \pm 10 \text{ pannels}$$

4

Also, simple cost analysis conducted. If necessary, charging the shuttle from the grid would only cost a few euros per charge (at Spain's current industrial rate of €0.15–0.20/kWh), which is significantly less than using diesel fuel for the same task. Long-term financial viability is demonstrated when the cost savings on gasoline eventually balance the solar panel investment.

- Energy - Battery Needs:

The shuttle's energy consumption was calculated using the "Ruta de los Volcanes" circuit. Topographical data—specifically variations in elevation—and a conservative driving cycle, including frequent stops and an average speed of approximately 20 km/h, were used to estimate battery usage per tour. This analysis included assessments of rolling resistance, gravitational force on inclines, and accessory loads such as cooling fans and the audio system. The resulting estimates indicated that a 20 kWh battery would require recharging after approximately two to three complete circuits, thereby influencing the selection of battery capacity.

Variables	Assumptions
Circuit Length	$\rightleftarrows$ 28 km
Average Speed	25 km/h
Vehicle Mass (Shuttle + Passengers)	2000 kg
Rolling Resistance Coefficient:	0.015
Air Drag Negligible	Due to low speed
Accessory Load	1.5 kW (Audio system, fans, lighting, etc.)
Battery Capacity	20 kWh
Total Elevation Gain	480 m

Table 6: Assumptions

Rolling Resistance Energy Consumption

Formula:

$$E_{rr} = m \times g \times C_{rr} \times d$$

$$m = 2000\text{ kg}$$

$$g = 9.81\text{ m/s}^2$$

$$C_{rr} = 0.015$$

$$d = 23 \times 1000 = 23000$$

Calculations:

$$E_{rr} = 2000 \times 9.81 \times 0.015 \times 23000 = 6.77 \times 10^6 J$$

1

Convert to kWh:

$$E_{rr} = \frac{3.6 \times 10^6}{6.77 \times 10^6} \approx 1.88 kWh$$

Gravitational Energy Consumption (Uphill)

Formula:

$$E_{grav} = m \times g \times h$$

Where:

- $h=480\text{ m}$

2

Calculations:

$$E_{grav} = 2000 \times 9.81 \times 480 = 9.42 \times 10^6\text{ J}$$

Convert to kWh:

$$E_{grav} = \frac{9.42 \times 10^6}{3.6 \times 10^6} \approx 2.62\text{ kWh}$$

Regenerative Braking Recovery:

Assume 60% recovery efficiency on downhill: Recovered energy $\approx 1.57\text{ kWh}$

Net gravitational energy consumption:

3

$$2.62 - 1.57 = 1.05\text{ kWh}$$

Accessory Load Energy Consumption

Average tour time:

$$\frac{28 \text{ km}}{25 \text{ km/h}} = 1.2 \text{ hours}$$

4

Accessory consumption per tour:

$$1.5 \text{ kW} \times 1.12 \text{ h} = 1.68 \text{ kWh}$$

This value (1.5 kW) is a common estimation for small electric shuttles or low-speed tourist vehicles, especially those operating at low speeds (~20 km/h) without heavy-duty air conditioning.

Rolling Resistance (1.88) + Gravitational Effort (2.62) + Accessory Load (1.68)

Battery Autonomy Estimation

5

Battery capacity = 20 kWh

Energy per tour = 6.18 kWh

Maximum theoretical number of full tours before charging:

$$\frac{20}{6.18} \approx 3.2 \text{ tours}$$

Total daily tours required: 84 tours per day

Total daily energy demand:

$$84 \times 6.18 \text{ kWh} = 519.12 \text{ kWh per day}$$

You will realistically operate 3.2 tours per charge (considering safety margin).

6

Battery cycles needed per day (per shuttle):

$$\frac{84}{3.2} = 27$$

Full battery charges per day

An effective way to guarantee the viability of running 84 trips a day was to implement a battery swapping system. Relying exclusively on traditional charging would cause major downtime and operational delays because the energy simulation indicated that each tour consumes roughly 6.18 kWh and the 20 kWh battery only permits three tours per charge. The shuttles can run continuously without long charging stops by putting in place a battery switching system that allows empty batteries to be quickly replaced with fully charged ones at the stations. The high-frequency schedule needed for the "Ruta de los Volcanes" circuit is supported by this method, which also increases operating efficiency and allows for continuous service.

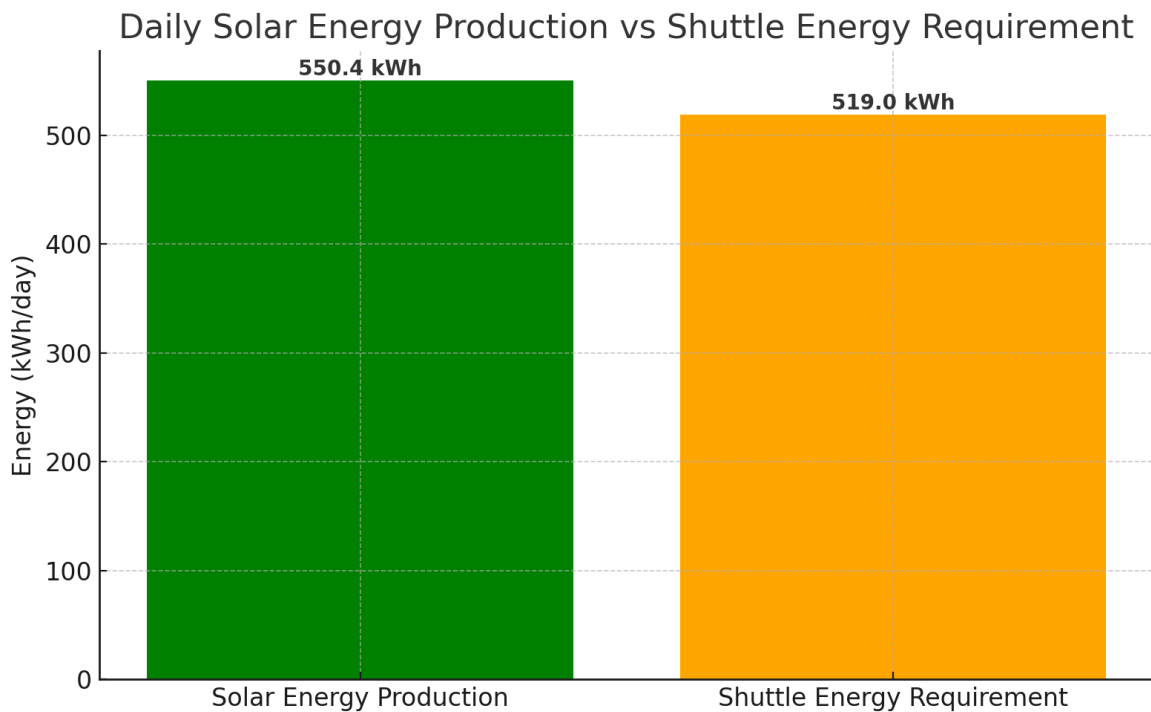


Figure 25: Daily Production - Requirement

5. DISCUSSIONS

This section presents an analysis of the findings and evaluates how effectively the project's outcomes align with the original goals and objectives. Potential advantages and challenges associated with implementing the proposed shuttle system in Timanfaya National Park are also considered, along with comparisons between the results and initial assumptions or established theoretical frameworks. Each of the primary objectives is examined individually, assessing the extent to which it was achieved and substantiated by the project's findings.

5.1 Achievement of Project Objectives

The results of the project show that each of the four main goals was accomplished:

1. *Environmental Protection*: By substituting solar-electric shuttles for diesel-powered buses, the suggested solution produces zero direct emissions. According to simulation results, a fleet of 20 shuttles could potentially reduce its CO₂ emissions by about 810,000 kg per year. This reflects successful case studies in Yosemite and Zion National Parks (Steinbach, 2023) and is consistent with sustainability goals outlined in UNEP regulations (UNEP, 2023).
2. *Improved Visitor Experience*: By including panoramic vistas, low noise levels, and an interpretive audio guide system, the shuttle design raises visitor satisfaction. These results are in line with tourism research that connects comfort and environmental immersion to visitor experience (UNWTO, 2022). Batteries have environmental costs throughout their life cycle, even though they are not direct emissions, therefore disposal and production effects could be a problem.

3. *Sustainability and Viability*: According to energy analysis, solar energy can supply the bulk of the daily energy consumption. Additionally, scalability and long-term operating viability are made possible by the system's modular design.
4. *Accessibility*: The shuttle's design incorporated universal elements including railings, wheelchair ramps, and low-floor access. This tackles the mobility obstacles noted in earlier research (Rodriguez et al., 2022).

5.2 Critical Reflection and Limitations

Despite achieving its stated goals, the project has a number of limitations that should be noted:

- **Simulation Assumptions**: Standard battery performance and typical weather circumstances served as the foundation for energy and capacity calculations. Performance may be impacted by real-world factors like bad weather, unforeseen visitor behavior, or equipment malfunctions.
- **Economic Analysis**: A thorough cost-benefit analysis is necessary for implementation, even though the financial viability of infrastructure investment was examined at a basic level.
- **Stakeholder Engagement**: Due to project scope limitations, formal stakeholder input (park management, visitor surveys) was not methodically gathered, even if informal feedback was mentioned.
- **Operational Factors**: Long-term vehicle durability, employee training, and maintenance needs were not thoroughly modelled.

5.3Linkage to Literature

The results of the project support previous research on the advantages of electric shuttle systems in protected areas for the environment and society (Gilleran et al., 2015; Steinbach, 2023). This project examined the particular difficulties of establishing sustainable transport on a small island biosphere reserve, in contrast to earlier research that mostly concentrated on major national parks in North America.

Additionally, adding universal design principles to ecotourism transport systems helps the lack of study on accessibility.

Objective	Achieved	Justification
Eliminate emissions	✓	Zero CO ₂ emissions using solar-powered shuttles
Improve visitor experience	✓	Enhanced tour with audio narration & open design
Ensure sustainability	✓	Solar energy supply, low running costs, scalable
Guarantee accessibility	✓	Ramp, wheelchair space, universal design

Table 7: Objectives Table

5.4 Risk Assessment

Table 6 presents a risk analysis that assesses the primary risks connected to the suggested system.

Risk	Impact	Likelihood	Mitigation
Weather dependency	Energy production reduced on cloudy days	Medium	Install surplus solar capacity & include energy storage
Technical failure	Shuttle or solar panel malfunction	Low	Apply regular maintenance schedule
High initial investment	Financial risk for stakeholders	Medium	Seek funding from sustainability grants
Visitor safety	Potential accidents during shuttle boarding	Low	Provide ramps, handrails, and staff assistance
Battery replacement problems	Time lost replacing shuttle batteries during operation	Medium	Use swappable battery system & schedule efficient battery swaps
Visitor Overcrowding	Increased waiting time during peak periods	Medium	Adjust fleet size and shuttle frequency

Table 8: Risk Table

5.5 Broader Implications

Timanfaya National Park would be positioned as a pioneer in environmentally friendly travel by putting this shuttle system into place, which would be in line with the principles of the UNESCO

Biosphere Reserve. Other protected places looking for low-impact mobility solutions can use the suggested methodology. However, accurate stakeholder participation, continuous maintenance, and budgetary planning will be necessary for successful implementation.

6. CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

This initiative has effectively realized its objective of creating a sustainable, alternate mode of transportation for the popular "Ruta de los Volcanes" trip in Timanfaya National Park. A solar-powered electric shuttle system that satisfies the park's requirements for accessibility, tourist satisfaction, and environmental preservation through a methodical design process has been created. The main problems with the current diesel bus system are addressed by the final design, which consists of a number of electric shuttles powered by solar charging stations. It completely removes direct carbon emissions, safeguarding the park's delicate volcanic ecosystem and air quality. By providing more intimate, calm, and instructive tours of the volcanoes, it improves the experience of visitors. Additionally, it guarantees that the trip is accessible, enabling visitors of all abilities to take in the beauty of the park. The suggested method would, in quantitative terms, cut CO₂ emissions by an estimated 100% (saving 70-80 tons of CO₂ annually) and probably boost visitor satisfaction (anticipated to improve by more than 20% due to the additional features and comfort). The shuttle network is built to be sustainable and effective while conforming to the park's preservation goals, fulfilling the project's goal of striking a balance between ecological protection and operational effectiveness.

Every one of the four project goals was completed:

- *Sustainability and viability*: achieved by using solar power, which ensures low operating costs and long-term feasibility.
- *Improved visitor experience*: Achieved by better vehicle design (panoramic views, interpretation).
- *Environmental protection*: Achieved by using renewable energy and zero-emission electric transportation.
- *Accessibility*: Achieved by incorporating universal design elements so all visitors can participate in the tour.

Timanfaya National Park might use the conceptual result as a comprehensive implementation plan. If implemented, this will highlight Timanfaya's status as a UNESCO Biosphere Reserve model and making it one of the first parks in Spain with a completely sustainable transit infrastructure.

6.2Future Work

Even though the project's design and planning goals were met, there are still a number of areas that could use improvement, particularly as it moves from design to possible implementation:

- Prototype Development and Testing:

Building or acquiring an electric shuttle prototype in accordance with the concept and testing it in real-world Timanfaya conditions is a natural next step. Assumptions regarding performance would be verified by real-world testing; for example, it would confirm that the shuttle can manage the steepest road parts and that the battery range is enough under tour conditions, such as high temperatures and a full passenger load. A small group of guests, including those with disabilities, should be invited to ride the shuttle and provide feedback on its comfort, accessibility, and audio guide material as part of the prototype

testing process. Before expanding the fleet, these comments can help make small design changes. Any unexpected problems with solar yield or charging times will also be found by testing the charging station on-site using the prototype.

- Optimization and Detailing:

Technical areas can be further optimized in future work. Investigating different battery technology options, such as LiFePO₄ batteries, which have longer lifespans and are more thermally stable (maybe advantageous in hot climates), could be one way to reduce the situation. Another area for optimization is solar energy storage; flexibility might be increased by installing a battery storage system at the station to preserve solar energy and charge shuttles after dark. A more thorough simulation including variability (high tourist days vs. off-peak season) could help us appropriately calculate the number of shuttles and charging infrastructure. We sized the system based on average utilization.

- Scalability and Transferability:

Future work might explore how this model can scale or transfer to other routes or parks. Timanfaya has one main tour route, but the concept could be extended to other visitor areas in Lanzarote (or other Canary Island parks). Engaging with other national park services to share the design and get feedback could also be valuable – their input might reveal additional considerations or features to add (such as integration with existing park apps or multi-modal links with public transport to the park).

- Monitoring and Ongoing Improvement:

After the system is put into place, it will be crucial to collect data and keep an eye on its functionality. Key performance indicators (KPIs) should be established in future work, such as actual emissions avoided, visitor satisfaction ratings from surveys, vehicle

uptime, etc. After a year of operation, an academic study might be conducted to formally assess the effects (this could be a research article on the environmental and social implications of Timanfaya's shuttle system). Continuous improvement may result from the findings. For instance, if surveys indicate that tourists prefer longer tours, the route might be expanded.

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Appendix A: Gantt Chart

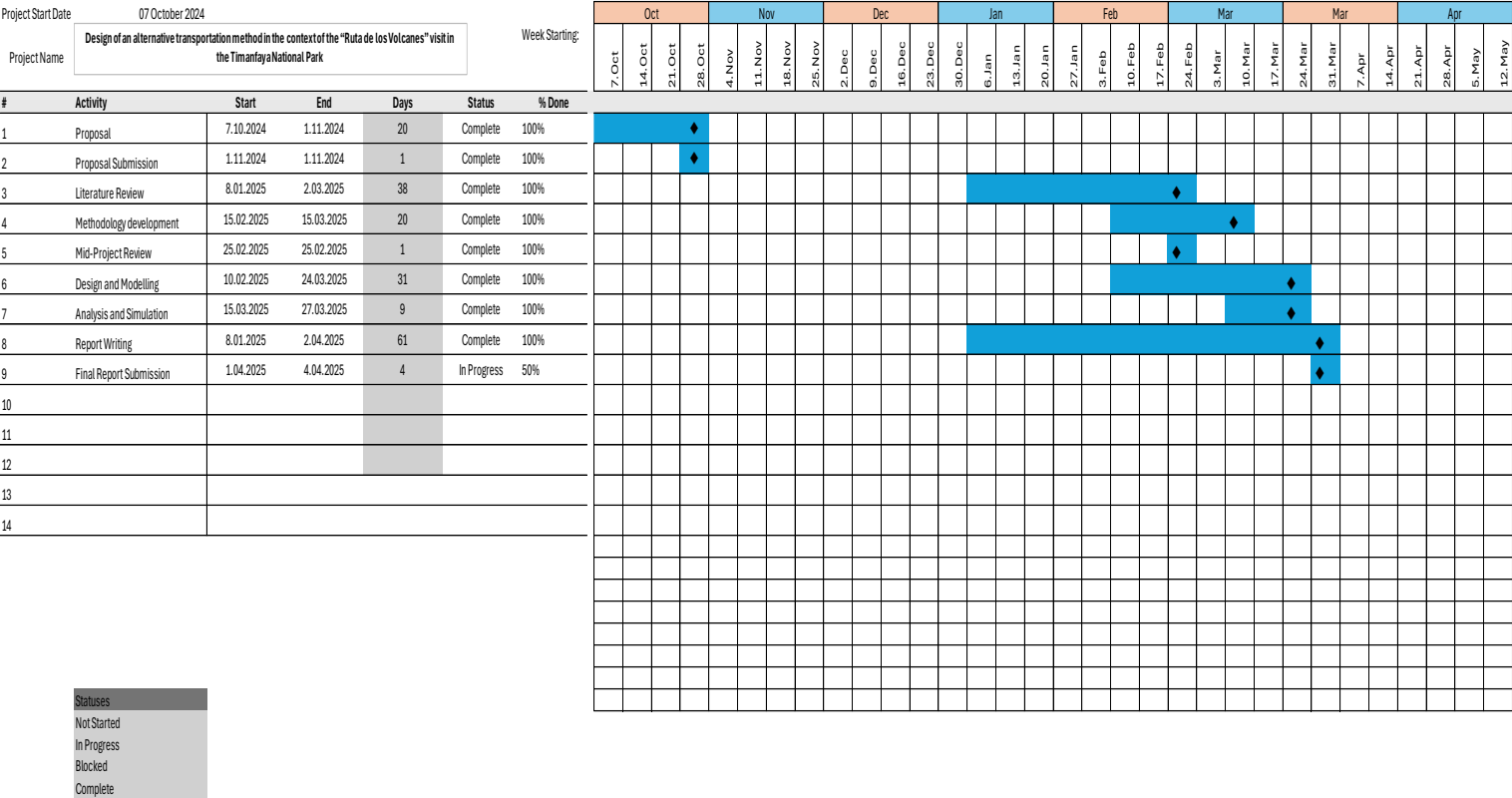


Figure 26: Gantt Chart



Figure 27: QR for Gantt Chart

Appendix A2: Meeting Block Tables

Meeting No.:	1	Meeting Date:	11/10/24
Student Comments:			
We discussed the reasoning behind my project selection and explored potential sources and references that could support the early stages of research.			
Supervisor Signature:	Mike Sharp		

Meeting No:	2	Meeting Date:	29/10/2024
Student Comments:			
We reviewed my project proposal and discussed which research methodologies would be appropriate. Mike advised me to align my methods with the double diamond process.			
Supervisor Signature:	Mike Sharp		

Meeting No:	3	Meeting Date:	19/02/2025
Student Comments:			
This informal session was mainly about the final report layout. Mike told me to feel free with the visual layout but reminded me to follow the template instructions to keep it professional. He also pointed out the importance of clear section transitions.			

Supervisor Signature:	Mike Sharp
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Meeting No:	4	Meeting Date:	14/03/2025
Student Comments:			
After showing Mike my early shuttle concept drawings during the Industry 4.0 lecture, he commented on the good direction but said I needed to work more on accessibility aspects and make the visuals cleaner.			
Supervisor Signature:	Mike Sharp		

Meeting No:	5	Meeting Date:	17/03/2025
Student Comments:			
I showed Mike the updated design following his feedback. He suggested finding out more about solar panel integration. He also mentioned I shouldn't add the informal survey.			
Supervisor Signature:	Mike Sharp		

Meeting No:	6	Meeting Date:	21/03/2025
Student Comments:			

Since the project was nearly complete, I asked Mike to check the progress and give final pointers. He also advised simplifying some of the text for the word count.

Supervisor Signature:	Mike Sharp
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Meeting No:	7	Meeting Date:	31/03/2025
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Student Comments:

We had a Teams call about the final version of the poster. Mike pointed out spacing issues and mentioned that I should balance the font sizes better.

Supervisor Signature:	Mike Sharp
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Meeting No:	8	Meeting Date:	01/04/2025
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Student Comments:

During Presentation Day, Mike told me he was pleased with the result but recommended a quick review of the report before final submission. We agreed to meet one more time for that purpose.

Supervisor Signature:	Mike Sharp
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Meeting No:	9	Meeting Date:	04/04/2025
Student Comments:			
This was the last session before submission. I sent Mike my final report, and he pointed out a few minor layout and wording adjustments which I applied right after the meeting.			
Supervisor Signature:	Mike Sharp		

Appendix B: Other Information

COST ANALYSIS

<i>Cost Component</i>	<i>Unit Cost (£)</i>	<i>Quantity</i>	<i>Total Cost (£)</i>
Electric Shuttle Vehicle	150,000	20 shuttles	3,000,000
Solar Panel Installation (2 m ² /panel)	250 per panel	167 panels	41,750
Battery Replacement (every 5 years)	12,000 per shuttle	20 shuttles	240,000

Charging Infrastructure	50,000	1 system	50,000
Maintenance & Staff Costs (Annual)	100,000	Per year (fixed)	100,000/year
Project Implementation & Logistics	250,000	One-time	250,000
Total Initial Investment	3,591,750		

Table 9: Cost Analysis

If the park charges, for example, £5 per shuttle ticket and serves 300,000 visitors annually:

Annual Revenue (Est.):

$$£5 \times 300,000 = £1,500,000$$

Annual Operational Cost:

$$£100,000$$

Payback Period:

$$\frac{\text{Initial Investment (£3,591,750)}}{\text{Annual Profit (£1,400,000)}} \approx 2.56 \text{ years}$$

MATERIAL SELECTION

Component	Selected Material	Key Properties	Reason for Selection
Chassis Frame	Aluminum Alloy (6061-T6)	Lightweight, high strength, corrosion resistance	Reduces vehicle weight, increases energy efficiency
Body Panels	Glass Fibre Reinforced Polymer (GFRP)	Low density, high durability, cost-effective	Lightweight, aesthetic flexibility, cost reduction
Windows	Laminated Safety Glass	Impact resistance, UV protection	Ensures passenger safety and visibility
Floor Structure	Recycled Thermoplastic Composite	Lightweight, recyclable, slip-resistant	Sustainability focus, safety, and weight saving
Handrails & Supports	Stainless Steel (Grade 304)	Corrosion resistance, strength	Hygiene, durability, visitor safety
Battery System	Lithium-ion (LFP)	High energy density, long cycle life	Reliability and extended range

Solar Panels	Monocrystalline Silicon	High efficiency (20%), long lifespan	Maximum energy output, compact installation
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Table 10: Material Selection Table

DESIGN OVERVIEW

Seats:

Design Overview:

- With a cushioned seating surface and backrest, the seat offers an ergonomic design.
- Both sides have armrests installed for the stability and comfort of the passengers.
- Visitors can store their bags or other personal items in the cutout base section of the seat structure.

Dimensions:

- Width: 0.50 m
- Depth: ~ 0.40 m
- Height: 1.20 m
- Armrest distance from side: 0.07 m

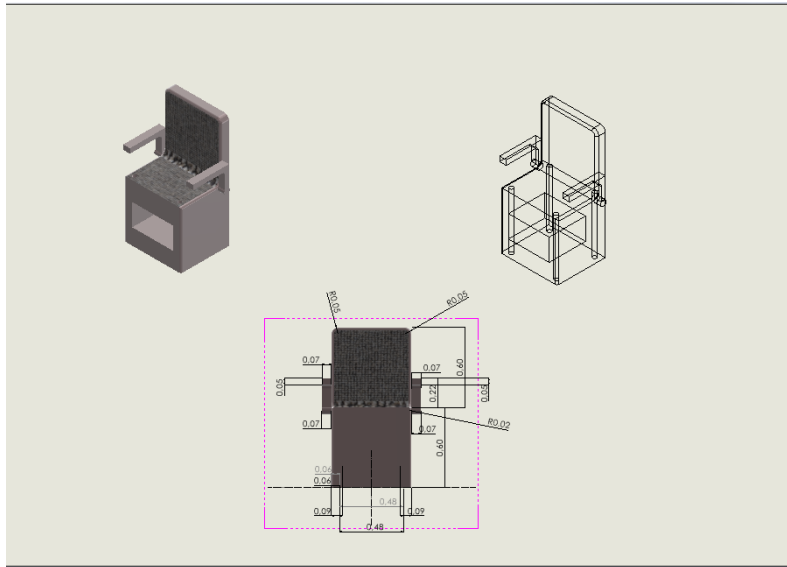


Figure 28: Seat Drawing

Audio Speaker:

Design Overview:

- Every shuttle has a multilingual audio system that is activated by GPS. The shuttle's onboard speakers automatically broadcast pre-recorded stories when it approaches significant landmarks, like the base of a volcano.
- Solar panels are used to charge the shuttle's auxiliary battery, which also powers the speaker system.
- This speaker is positioned in the centre of the vehicle on the upper wall or ceiling, as seen in the complete shuttle view.
- Improves visitors' educational experience by offering live or recorded tour guides.

Dimensions:

- Width: 0.2 m
- Depth: 0.3 m
- Height: 0.7 m

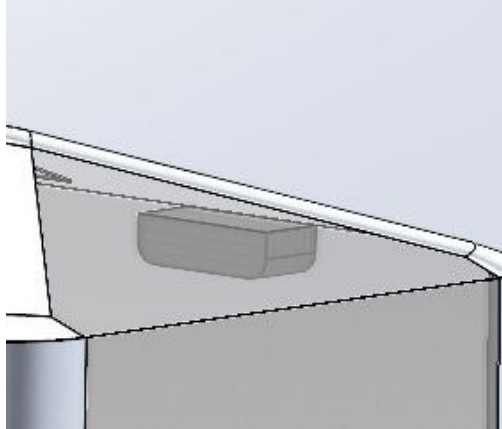


Figure 29: Speaker photo

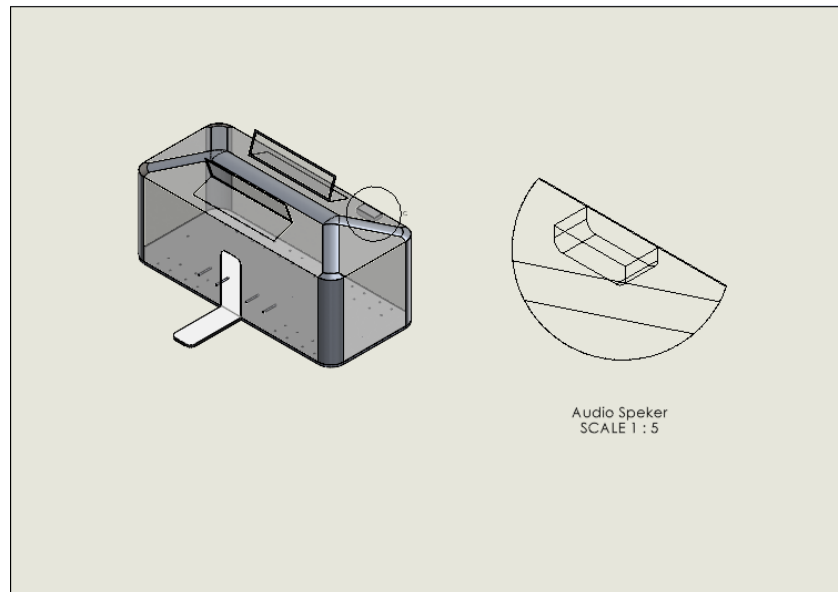


Figure 30: Speaker Drawing

Handrails:

Design Overview:

- To provide safer trips, the railings are added to the design.

- They are positioned next to seats and entrances, as the vehicle layout illustrates, in accordance with universal design guidelines for public transportation.
- Two handrails are positioned next to the windows to provide visitors with an improved experience. For instance, they can safely hold handrails and enjoy the tour when they wish to move around the shuttle.
- There are two more railings in the center. They can use that when they are leaving or coming into the vehicle to prevent any issues when the shuttle takes a break or begins to move.

Dimensions:

- Height: ~ 2.60 m
- Width: 0.04 m
- Depth: 0.03 m

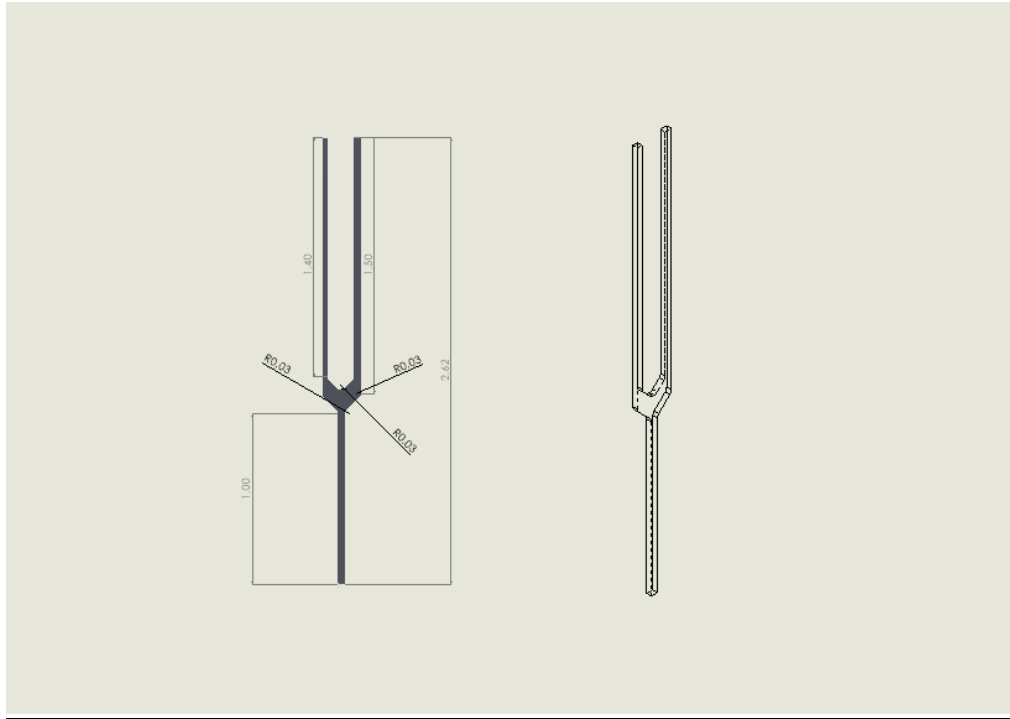


Figure 31: Handrail Drawing

Hand bars for Wheelchairs:

Design Overview:

- To provide safer trips for wheelchair users, the hand bars are implemented next to the doors
- The distance between two wheelchair supporter bar is 70 cm which is going to be enough for any guest with special needs.

Dimensions:

- Diameter: 0.025 m
- Height: 0.7 m
- Depth: 0.5 m

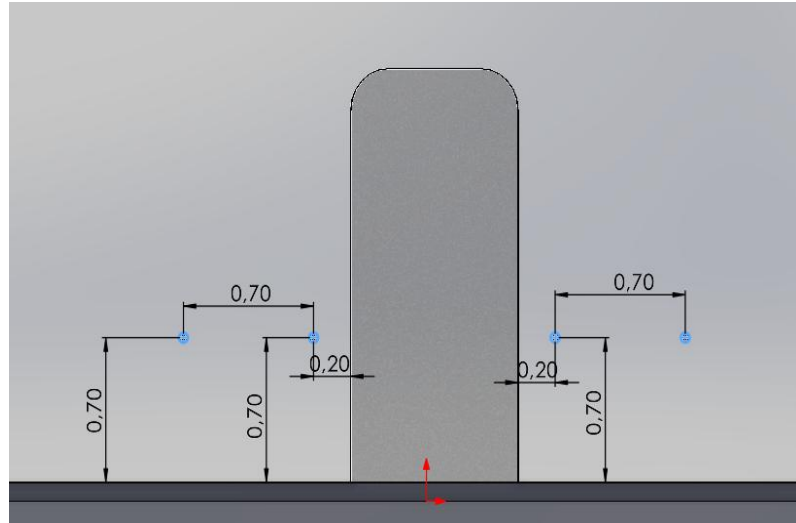


Figure 32: Hand supporter bars for wheelchair

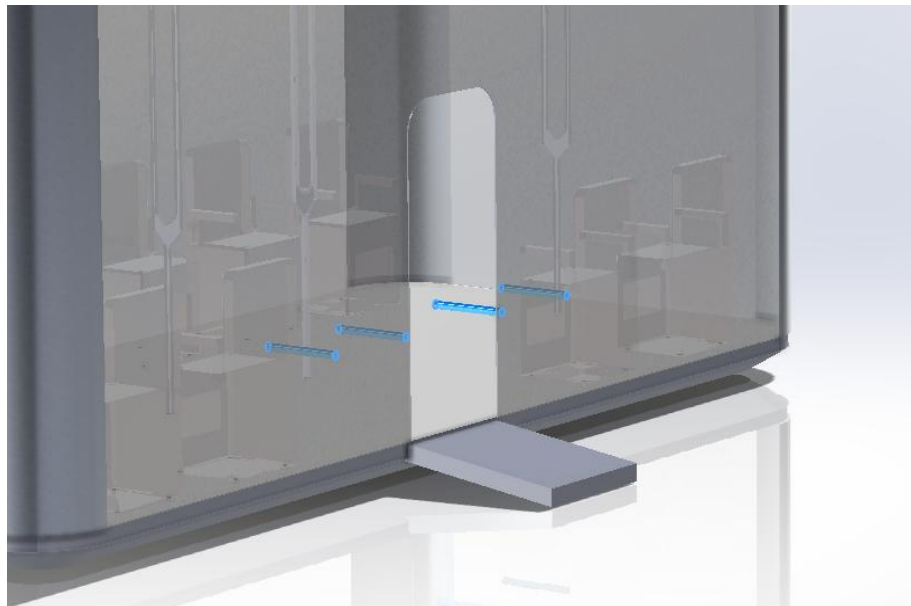


Figure 33: Hand supporter for wheelchair 2

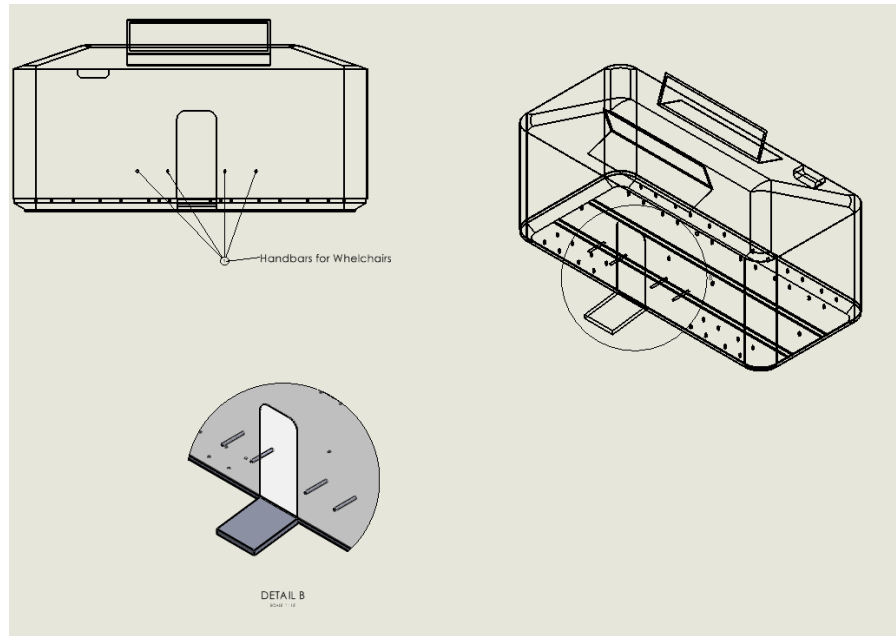


Figure 34: Hand supporter bars drawing

Ramp for Wheelchair

Design Overview:

- Wheelchair users and tourists with reduced mobility will benefit from the shuttle's special accessible ramp.

Dimensions:

- Ramp angle: 172° from the vertical surface ($\sim 8^\circ$ incline) — safe and optimum for standard wheelchair access.
- Height: 1 m
- Width: 0.9 m
- Depth: 0.3 m

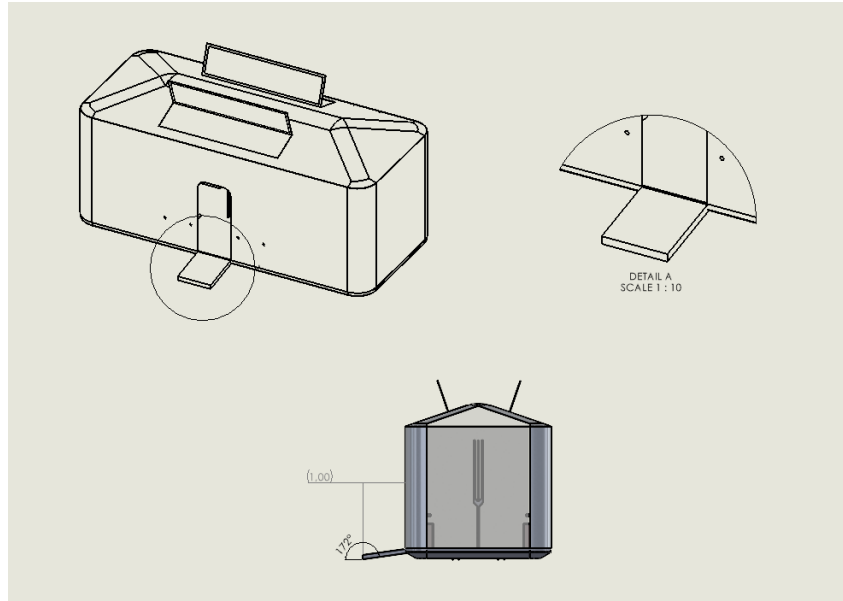


Figure 35: Accessibility Ramp Drawing

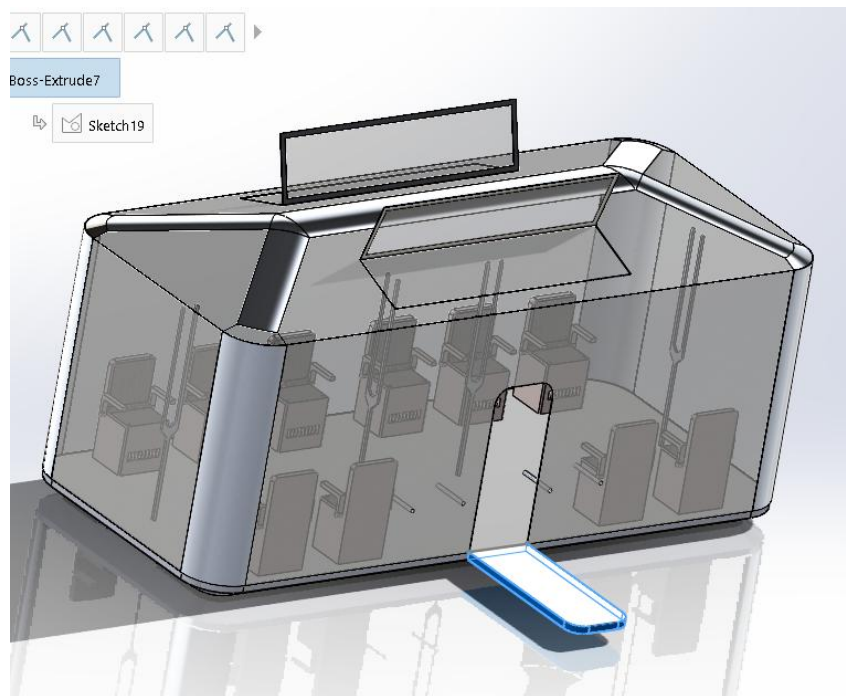


Figure 36: Accessibility Ramp 1

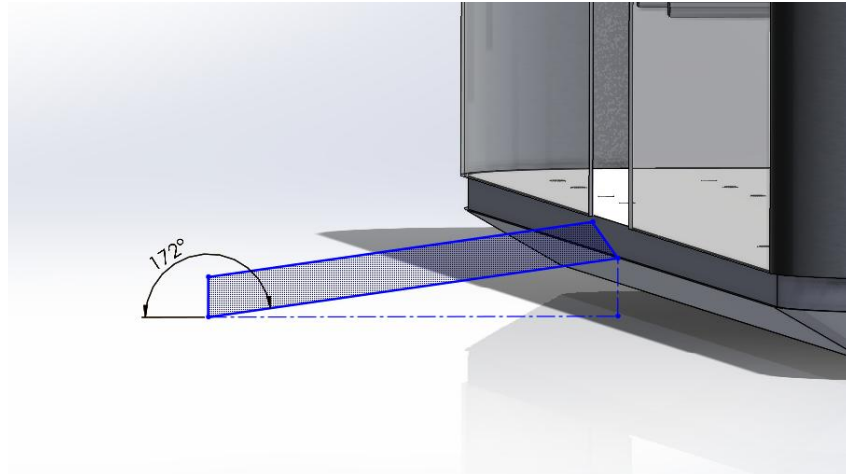


Figure 37: Accessibility Ramp 2

Solar Panels:

Design Overview:

- As calculated in the next section the number of solar panels needed is 167.
- For the benefit of the guests, the solar panels are positioned between two tramways. In essence, it is covered between the railway to avoid disturbing with the breathtaking views.
- We decided to use a system which is going to take the energy directly from the railway with the system explained in the previous section.
- The design is based on the regular solar panel requirements which can produce around 400W in an hour with Lanzarote's solar irradiance

Dimensions:

- Height: 2 m
- Width: 0.95 m
- Depth: 0.2 m

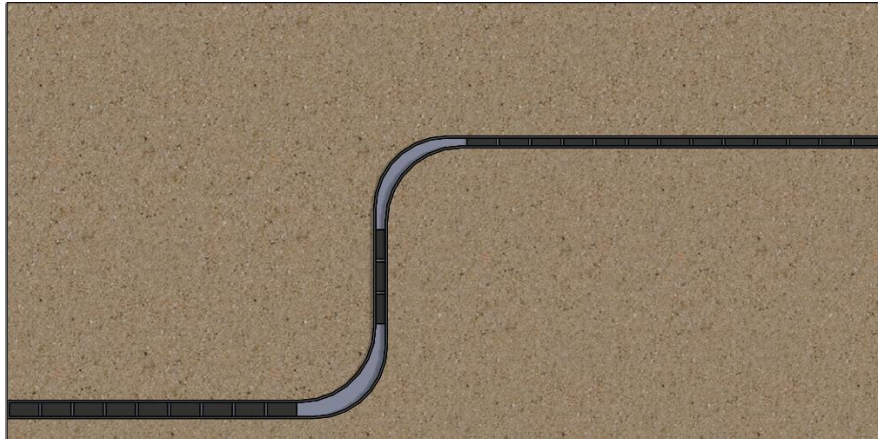


Figure 38: Solar Panel 1

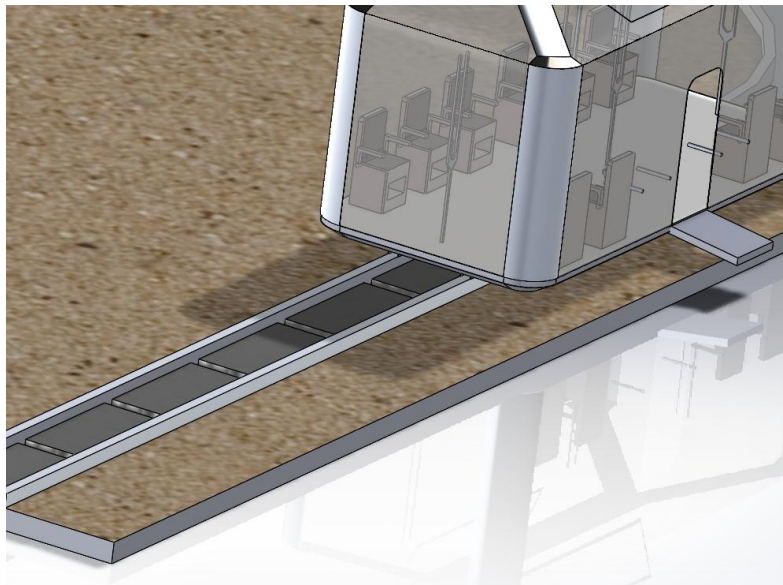


Figure 39: Solar Panel 2

OBJECTIVES

Environmentally Friendly System

The design has successfully achieved the main goal of removing direct emissions. By switching from diesel to electric, the route's CO₂ and other pollution emissions would be completely eliminated, significantly improving Timanfaya's ecology and air quality. The indirect emissions

are important subject of discussion since, although the shuttles don't emit any exhaust, the upstream emissions from electricity generating can. In theory, our system achieves near-zero overall emissions by mitigating this using solar power. Even if some grid electricity is periodically used, renewable energy sources are rapidly supplying Spain's system, thus the total amount of power required is little.

This result shows how local adjustments (park shuttles) support global sustainability goals and is consistent with climate action policies (UNEP, 2023). Notably, there is also a significant decrease in noise pollution because electric motors are quite quiet, which means that the natural soundscape—which includes wind, geothermal vents, and wildlife—will be less disrupted.

Batteries have environmental costs throughout their life cycle, even though they are not direct emissions, therefore disposal and production effects could be a problem. A strategy for recycling or reusing batteries after about eight to ten years should be part of any future implementation.

Visitor Experience Enhancements

The new shuttles promise increased educational engagement (audio tours) and immersive viewing (open sides, ability to stop at locations of interest). Visitors tend to be mostly passive under the current arrangement. They become more involved with the suggested system, such as when they learn about the formation of each crater as they go by it or experience the temperature and breeze variations in different parts of the lava field. Satisfaction should rise as a result of these experiential enhancements. One factor to take into account is capacity; because each shuttle is smaller, guests may have to wait a few more minutes for the next one during busy periods. However it is way better than squeezing 50 people into a large bus.

Sustainability and Operational Feasibility

According to the energy study, nearly all of the demand can be satisfied by solar input. As a result, everyday operations become substantially independent of external fuel or electricity, creating a sustainable closed-loop. In terms of economics, after initial expenses (vehicles and solar infrastructure), operating costs are minimal (EV maintenance is easier and solar energy is free). Scalability is another aspect of sustainability: the system can start with a small fleet and grow if necessary. It can also adjust to future technological advancements (e.g., these shuttles might be upgraded if autonomous driving becomes feasible or older batteries can be replaced as battery technology advances). The complexity of charging and weather dependence are two practical problems to draw attention to. A hybrid solar/grid system is sensible because backup charging from the grid or a generator would be required to sustain service in the event of an unusually high number of cloudy days. The extreme volcanic dust environment may also damage the cleanliness of solar panels or electrical connections, even though EVs have fewer moving parts (Contese Agency, 2024).

Accessibility and Inclusion

The shuttle system would significantly increase accessibility by including a ramp, wheelchair area, and other simple access features. There are currently obstacles for tourists with mobility limitations (limited space on buses). Wheelchair users can get onboard with assistance and enjoy the same tour experience as everyone else as a result of the new technology, which reduces such barriers. As mentioned in our aims, this satisfies the objective of "guaranteeing accessibility for all guests." For example, the low-floor design will also be helpful for elderly visitors or children, who will find it easier to board than climbing bus steps. More space is provided by the open layout, which lessens the stress of crowds. The only warning is to make sure the equipment and ramp are durable; they need to be tested for dependability because any

failure (such as a stuck ramp) could cause service interruptions. All things considered, the project's accessibility discussion is excellent: the goal is covered in detail.

COMPARISON WITH EXISTING SYSTEM

CRITERIA	CURRENT DIESEL BUS	PROPOSED SOLAR-ELECTRIC SHUTTLE
CO ₂ EMISSIONS	27 kg/tour	0 kg/tour
NOISE POLLUTION	High	Minimal
ACCESSIBILITY	Limited (no wheelchair access)	Full compliance
OPERATIONAL COST	High (fuel, maintenance)	Low (solar, minimal maintenance)
VISITOR EXPERIENCE	Standard	Enhanced (quiet, scenic, informative)

Table 11: Comparison Table